

20. Mammoth is the ancestor of-

- (a) Dog (b) Horse
(c) Camel (d) Elephant

R.A.S./R.T.S. (Pre) 2008

Ans. (d)

Extinct mammoth is more closely related to today's African elephant or Asian elephant. Modern elephants and woolly mammoths share a common ancestor, that lived about 6 million years ago.

21. Dinosaurs were :

- (a) Cenozoic reptiles
(b) Mesozoic birds
(c) Paleozoic amphibians
(d) Mesozoic reptiles

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (d)

The Mesozoic Era is the age of the dinosaurs and lasted almost 180 million years from approximately 250 to 65 million years ago. This era includes three well known periods, called the Triassic, Jurassic and Cretaceous periods.

22. Dinosaurs were in-

- (a) Paleozoic era (b) Cenozoic era
(c) Mesozoic era (d) Pleistocene era

U.P. Lower Sub. (Pre) 1998

Ans. (c)

See the explanation of above question.

23. With reference to the evolution of living organisms, which one of the following sequences is correct?

- (a) Octopus–Dolphin–Shark
(b) Pangolin–Tortoise–Hawk
(c) Salamander–Python–Kangaroo
(d) Frog–Crab–Prawn

I.A.S. (Pre) 2009

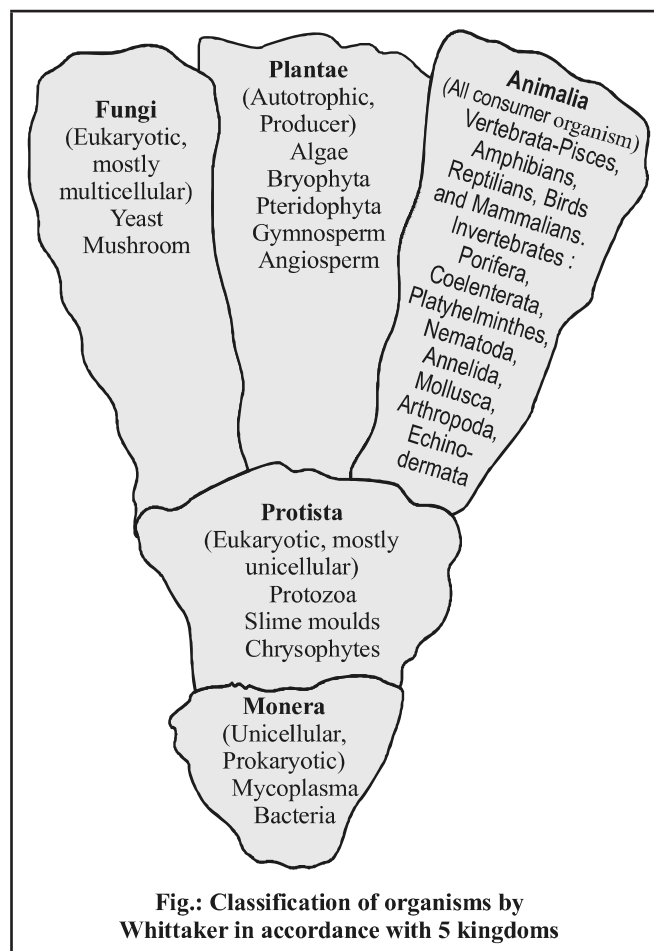
Ans. (c)

Salamander is a tailed amphibian which appeared about 360 million years ago, Python is a reptile which appeared about 300 million years ago while Kangaroo is a pouched mammal or marsupial which appeared about 145 million years ago. With reference to the question, the correct sequence is 'Salamander–Python–Kangaroo'.

Taxonomy

Notes

- Taxonomy is a system for naming and organizing things especially plants and animals, into groups that share similar qualities.
- Carl (Carolus) Linnaeus** (1707-1778) was a Swedish botanist, physician and zoologist who formalized **binomial nomenclature** – the modern system of naming organism. He is known as the '**father of modern taxonomy**'.
He classified the organisms into five kingdoms viz. **Monera, Protista, Fungi, Plantae** and **Animalia**.



Kingdom Monera :

- Kingdom Monera includes organisms that are single celled. The microorganisms are considered as the most ancient living forms on the earth. All the organisms of this kingdom are **prokaryotes**. These cells do not have nuclear membrane. The kingdom Monera includes

bacteria, cyanobacteria and mycoplasma. They do not have specific mode of nutrition. They can be either **aerobic** or **anaerobic**. These organisms have rigid cell wall which is made up of **peptidoglycan**. Cell organelles like endoplasmic reticulum, mitochondria are absent. Reproduction is by spore formation or binary fission.

- **Mycoplasma** are known to be the smallest living cells. They completely lack cell wall. This characteristic makes them naturally resistant to many common antibiotics such as penicillin or other beta-lactum antibiotics that target cell wall synthesis. They can survive without oxygen. Most of the mycoplasma are pathogenic in animal and plants e.g. **M. pneumoniae** causes atypical pneumonia in humans.
- **Cyanobacteria** are photosynthetic, that is, they can synthesize their own food. They are the photosynthetic prokaryotes able to produce oxygen. Cyanobacteria are also called 'blue-green algae', though the term algae in modern usage is restricted to eukaryotes. They can be found in almost every terrestrial and aquatic habitat. Cyanobacteria such as *Anabaena* (a symbiont of the aquatic fern *Azolla*) can provide rice plantation with biofertilizer.
- **Bacteria** are microscopic, single celled primitive organisms. Generally these are about 2 to 4 micron in length. They can live within soil, in the ocean and inside the human gut. Human relationship with bacteria is complex. Sometimes they lend a helping hand by curdling milk into yogurt or helping with our digestion. At other times they are destructive, causing diseases like cholera, syphilis, leprosy, tuberculosis and MRSA (**Methicillin-resistant Staphylococcus aureus** – a type of staph bacterial infection that is hard to treat because it cannot be killed by many common antibiotics).
- On the basis of their nutrition, bacteria have been classified into **heterotrophic** and **autotrophic** bacteria.
- Autotrophic bacteria are of two types – (i) **Photosynthetic** and (ii) **Chemosynthetic**.
- **Archaeobacteria** belong to the domain Archaea and are single celled organisms that tend to live in extreme environments, like hot springs or high salt regions. In fact, these organisms boldly grow there, where others can not. They have been called **Extremophiles**, which means lover of extremes. Example – **Methanogens** are microorganisms that produce **methane** as a metabolic by product in hypoxic condition. They are common in wetland, where they are responsible for **marsh gas** and in the digestive tract of animals such as ruminants and human, where they are responsible for the methane content of belching (emit wind noisily from the stomach through

the mouth) in ruminants and flatulence accumulation of gas in alimentary canal in humans. These microorganisms are similar to bacteria in size and simplicity of structure but radically different in molecular organization. They are now believed to constitute an ancient intermediate group between the bacteria and eukaryotes. They are also called **archaea**.

- **Rhizobia** are diazotrophic bacteria that fix nitrogen after becoming established inside the root nodules of legumes. Rhizobia are unique in that they are the only nitrogen fixing bacteria living in a **symbiotic** relationship with legumes.
- Unicellular organisms reproduce generally by asexual means. Asexual modes of reproduction includes binary fission, multiple fission, fragmentation, budding etc.

Kingdom Protista :

- All single celled eukaryotes are placed under Protista. They are organisms which are unicellular or unicellular - colonial and which form no tissues. Protista kingdom includes chrysophytes (diatoms and golden algae), dinoflagellates, euglenoz, slime moulds and protozoas such as amoeba, paramecium, flagellata, ciliophora etc. Some protists reproduce sexually using gametes, while others reproduce asexually by binary fission.
- Protists live in water, in moist terrestrial habitats, and as parasites and other symbionts in the bodies of multicellular eukaryotes. Protists are almost certainly polyphyletic and they do not have an exclusive common ancestor.
- Some protists are significant parasite pathogens of animals (e.g. Plasmodium causes malaria in humans) and plants (e.g. Phytophthora infestans causes late blight in potatoes). Protist pathogens share many metabolic pathways with their eukaryotic hosts, which makes therapeutic target development extremely difficult – a drug that harms a protist parasite is also likely to harm its host.

Kingdom Fungi :

- Fungi is the plural word for fungus. Fungi are eukaryotic organisms. The study of fungi is called **Mycology**. Fungi lack chlorophyll and vascular bundle. Fungi are **heterotrophs**; they acquire food by absorbing dissolved molecules, typically by secreting digestive enzymes into their environment. Fungi reproduce both sexually and asexually.
- Yeasts, molds, mushrooms are the example of fungi.
- Yeast belongs to the division **Ascomycetes** of the kingdom Fungi. Most yeasts reproduce asexually by mitosis, and many do so by the asymmetric division process known as budding.

- Mold or mould grows in the form of multicellular filament called hyphae. In contrast, yeasts adopt a single-cell growth habit. Molds cause biodegradation of natural materials, which can be unwanted when it becomes food spoilage or damage to furniture etc.
- Mushrooms belongs to the division **Basidiomycetes** of the kingdom fungi. They are fleshy, spore-bearing fruiting body of a fungus, typically produced above ground on soil or on its food source. Mushrooms are used extensively in cooking, however, many mushroom species (e.g. *Amanita phalloides* which is called death mushroom) can be toxic.
- **Dimorphic fungi** are fungi that can exist in the form of both mold and yeast (e.g. *Penicillium marneffeii*, a human pathogen that grows as a mold at room temperature and as a yeast at human body temperature).
- A **mycorrhiza** is a symbiotic association between a fungus and a plant.
- A **lichen** is a composite organism that arises from the algae or cyanobacteria living among filaments of multiple fungi species in a mutualistic relationship.
- Adaptations of xerophytes include reduced permeability of the epidermal layer, stomata and cuticle to maintain optimal amount of water in the tissues by reducing transpiration, adaptations of the root system to acquire water from deep underground sources or directly from humid atmosphere and succulence, or storage of water in swollen stems, leaves or root tissues. Their leaves are modified into spines.
- **Phreatophyte** is a deep rooted plant that obtains significant portion of water that it needs from phreatic zone (zone of saturation) or the capillary fringe above the phreatic zone. The roots of such plants are approximately 25 to 30 meters reaching to underground water.
- **Hydrophytes** : Hydrophytes are also referred as aquatic plants or **macrophytes**. These plants require special adaptations for living submerged in water or at the water's surface. The most common adaptation is **aerenchyma**.
- **Halophytes** : A halophyte is a salt-tolerant plant that grows in waters and soils of high salinity, coming into contact with saline water through its roots or by salt spray, such as in saline semi-deserts, mangroove swamps, marshes and sloughs and seashores.
- **Epiphytes** : Epiphyte is a plant that grows on the surface of a plant and derives its moisture and nutrients from the air, rain, water (in marine environments) or from debris accumulating around it (Example – Orchids).
- A **flower** sometimes known as a **bloom** or **blossom**, is the reproductive organ found in flowering plants. The biological function of flower is to effect reproduction, usually by providing a mechanism for the union of male gamete with female gamete. Flowers may facilitate outcrossing or allow selfing.
- Ginger, potato, Garlic and Suran (jimikand) are the examples of **rhizome**, **tuber**, **bulb** and **corm** respectively.
- Shakarkand (Sweet-potato) belongs to the family **Convolvulaceae**. Its edible part is root.

Kingdom Plantae :

- These living organisms are made of eukaryotic cells and are multicellular. The cells have a cell wall which is made of cellulose. These are autotrophic and synthesize food by photosynthesis due to the presence of chloroplasts. They are divided into following divisions – Algae, Bryophyta, Pteridophyta, Gymnosperm and Angiosperm.
- **Pitcher Plant (Nepenthes)** : This plant is able to synthesise its own food. This plant is **insectivorous**. Pitcher plant grows in soil which lacks in nitrogen. To meet the deficiency of nitrogen, pitcher plant traps the insects.
- **Xerophytic plants** : A xerophyte is a species of plant that has adaptations to survive in an environment with little liquid water, such as desert or an ice or snow covered region. Popular examples of xerophytes are cacti, calotropis, aloe, pineapple and some gymnosperm plants.

Plant Taxonomy

Plant taxonomy is the science of classifying, naming, and identifying plants, organizing them into groups based on shared characteristics and evolutionary relationships. Plant taxonomy uses a hierarchical system to classify plants, with each level representing a group of organisms with shared characteristics :

- **Kingdom** : The broadest level, encompassing all plants (Plantae).
- **Phylum** : A major group within the kingdom, such as flowering plants (Magnoliophyta) or non-flowering plants (Bryophyta).
- **Class** : A group within a phylum, such as monocots or dicots (within Magnoliophyta)
- **Order** : A group within a class, such as the order Rosales (within Magnoliopsida).
- **Family** : A group within an order, such as the rose family (Rosaceae).
- **Genus** : A group within a family, such as the genus *Rosa* (roses).
- **Species** : The most specific level, representing a group of plants that can interbreed and produce fertile offspring, such as *Rosa rubiginosa* (dog rose).

- Sugarcane, potato and ginger are the modified stems. It means their food storage organ are stem.
- In shaljam (Turnip), carrot and sweet potato, storage organ are the modified roots.
- Pea is a dicotyledonous annual herbaceous plant. **Tendrils** are present to support the plant. It belongs to **Leguminosae** family.
- Sugarcane (**Saccharum officinarum**) belongs to the family **Graminae** or **Poaceae**. Generally **stem cutting** is used for its vegetative propagation. It has **nodes** and **internodes**.
- **Cloves** are the aromatic flower buds of a tree in the family Myrtaceae **Syzygium aromaticum**, native of Indonesia and commonly used as spice.
- Cloves health benefits include improving digestion, fighting bacteria, protecting the liver, fighting lung cancer, regulating blood sugar and relieving tooth pain.
- **Saffron** : Saffron is a spice derived from the flower of **Crocus sativus** (family - Iridaceae). The vivid crimson stamens and styles, called threads, are collected and dried to be used mainly as a seasoning and colouring agent in food. It has long been the world's most costly spice by weight. In saffron **safranal** and **picrocrocin** chemicals are found.
- **Okra (Ladies' finger)** : It is a flowering plant. It is valued for its edible green seed pods. Its pod is known as **capsule**. Its botanical name is **Abelmoschus esculentus**.
- **Turmeric (Curcuma longa)** is a flowering plant of ginger family-**Zingiberaceae**. The modified shoot (stem) rhizome is edible part of it. Turmeric contains **curcumin**, a substance with powerful anti-inflammatory and antioxidant properties.
- **Sorosis** : This type of fruit is found in Mulberry, Pineapple and Jack fruit. These fruits are derived from catkin, spike and spadix type of inflorescence.
- **Nut** : A nut is a fruit composed of an inedible hard shell and a seed, which is generally edible. The fruits of Cashew nut and Trapa (Singhara) are examples of nut. The edible part of Singhara is seed.
- **Lychee** : It is the sole member of the genus Litchi in the soapberry family, Sapindaceae. Fleshy aril of lychee is the edible part. It is a **drupe** (like a plum, a cherry or a mango), externally covered by a pink-red rough textured rind, easily removable.
- **Pome** : A fruit consisting of a fleshy enlarged receptacle and a tough core containing the seeds, e.g. an apple or pear.
- **Pepo** : Fruits having a fleshy, many-seeded interior and a hard or firm rind. The fruits of melon, squash and cucumber are called pepo.

- **Producers** : Producers are autotrophic, make their food by taking sunlight and using the energy to make sugar. The green plants are **producers**.
- **Consumers** : Consumers have to feed on producers or other consumers. Consumers are heterotrophs.
- **Herbivores** : Herbivorous animals are primary consumers.
- **Carnivores** : Secondary consumers which feed on herbivores (primary consumers).
- **Decomposer** : Decomposers are the garbage men of animal kingdom, they take all the dead animals and plants and break them down into their nutrient components so that plants can use them to make more food. These are **saprophytic**.

- **Retting** is a process, employing the action of micro-organisms, and moisture on plants to dissolve or rot away much of the cellular tissues and pectins surrounding bast-fibre bundles, and so facilitating separation of fibre from the stem of **sunhemp** and **jute**.
- The botanical name of papaya is **Carica papaya** (family- Caricaceae). Enzyme **papain** found in papaya which helps in digestion of protein. Papaya are yellow due to presence of xanthophyll pigments i.e. carixanthin present in the plastid of the fruit pulp.
- Red apples get their colour from **anthocyanin** pigment.
- The pigments found in some other edible parts of plants are – **carotene** in carrot, **lycopene** in tomato, **xanthophyll** in turmeric and **betanin** in Beta vulgaris (chukandar).
- The distinctive smell of garlic and onion is due to the presence of **sulphur** containing chemicals.
- Some compounds in onions, garlic or both, can be responsible for bad breath and even body odour. These include-
- **Allicin** – When the insides of a garlic bulb are exposed to air, a substance called alliin turns into allicin, which then changes into several sulphur containing compounds that gives garlic its smell.
- **Allyl methyl sulphide** – This compound is released from both garlic and onion, when they are cut. Once eaten, the substance is absorbed into blood stream, and emitted through the lungs and skin pores.
- **Cysteine sulfoxide** – This sulphuric compound in garlic and onions causes an unpleasant odour on the breath almost immediately after the vegetables are eaten.
- Garlic and onions add flavour to meals and can provide health benefits. Unfortunately, both can also cause bad breath, known as **halitosis**, especially when eaten raw.
- **Sulfenic Acid** found in onion irritates lacrymal glands and produces tears in eyes.

- **Capsaicin** is responsible for sharp taste in chili peppers. Its chemical formula is $C_{18}H_{27}NO_3$. When it comes in contact with skin or mucous membranes it produces a burning sensation. The amount of capsaicin in a specific species of peppers is measured using the **Scoville scale**.
- In botany, a **bud** is an undeveloped or embryonic shoot normally occurs in the axil of a leaf and stem or at the tip of stem. Once formed, a bud may remain for sometime in dormant condition or it may form shoot immediately. The term bud is also used in **Zoology**, where it refers to an outgrowth from the body which can develop into a new individual.
- **Cork** is obtained from the bark of **oak** which botanical name is **Quercus suber**. Its native land is mediterranean region.

Kingdom Animalia :

- Animalia kingdom includes all invertebrates and vertebrates animals. They are multicellular eukaryotic organisms.
- With few exceptions, animals consume organic material, breathe oxygen, are able to move and can reproduce sexually.
- Over 1.5 million living animal species have been described - of which more than 1 million are insects - but it has been estimated there are over 7 million animal species in total.
- **Arthropoda** is the largest phylum in animal kingdom. It includes species in all habitats which constitute 60% of all known species of animals.
- Crabs belong to the class **Malacostraca** (Phylum-Arthropoda).
- Class **Arachnida** (Phylum-Arthropoda) are characterized by having two body regions, a **cephalothorax** and an **abdomen**. They also have 6 pairs of appendages : 4 pairs of legs and 2 pairs of mouth part appendages, the first pair of mouth part appendages is called **chelicerae** and second pair is called **pedipalps**. Examples : Scorpions, Spiders.
- The **ticks** and **mites** belong to the order **Araneae** of class **Arachnida**. Their appendages are jointed, body is bisymmetrical and triploblastic.
- Class **Insecta** (Phylum-Arthropoda) encompasses all insects. Insects have three main body segments, the **head**, **thorax** and **abdomen**. They have 3 pairs of legs.
- The glowworm **Arachnocampa luminosa** (insect) is famous for having blue-green light on the end of its tail. Both adult and larva produce it in a process called **bioluminescence**. The glow is due to presence of an enzyme **luciferase**.
- In **female mosquitoes**, all mouth parts are elongated. The **labium** encloses all other mouth parts like a sheath. The **labrum** forms the main feeding tube, through which blood is sucked. Paired mandibles and maxillae are present, together forming the stylet, which is used to pierce an animal's skin. Thus the female mouth parts are adopted to suck and pierce.
- Male mosquitoes mouth is adopted to suck the nectar of flowers.
- The **trachea** is the respiratory organs and the **malpighian tubules** are excretory and osmoregulatory organs of the insects.
- **Echinoderm** is the common name given to any member of the phylum **Echinodermata** (hard, spiny covering or skin) of marine animals. The adults are recognisable by their (usually five points) radial symmetry and include such well known animals as **starfish**, **sea urchins**, **sand dollars**, **sea cucumbers**, **sea lilies** etc. Echinoderms are second-largest grouping of deuterostomes (a superphylum) after the chordates.
- Echinoderms are generally **oviparous** but some of them are **viviparous**.
- **Octopus** is a soft bodied, eight armed mollusc of the Phylum **Mollusca** class Cephalopoda. It is also known as **Devilfish**.
- Giant squids (**Architeuthis dux**) and colossal squid (**Mesonychoteuthis hamiltoni**) are the **largest invertebrates**. The length of colossal squids is less but weight is more than giant squids.
- **Homeothermic** or warm blooded animal species maintain a stable body temperature by regulating metabolic processes. The only known living homeotherms are **birds** and **mammals** (Their body temperature remains the same when it's cold or hot outside).
- **Heterothermic** or cold blooded animals, like reptiles, amphibians and fish become hotter and cooler, depending on the temperature outside (Their body temperature depends on whether it's cold or hot outside).
- **Nocturnal** animals can hunt, mate or generally active after dark. They have highly developed sense of hearing, smell and specially adapted eyesight. Examples of nocturnal animals are mosquito, bat, owl and kiwi. The aye-aye (**Daubentonia medagascariensis**), a type of lemur, is the world's largest nocturnal primate. It uses echolocation to find its prey—the only primate known to do so.
- Some animals name resemble with fish but they are actually not fishes e.g. **Jellyfish** (Cnidaria); **Starfish** (Echinodermata); **Silverfish** (Insecta) **Cuttlefish**

(Mollusca); **Hagfish** (Cyclostomata); **Devilfish** (Mollusca) and **Crayfish** (Arthropoda) etc.

- The some true fishes are flying fish, catfish, pipe fish, paddle fish, gold fish, globe fish, dogfish etc.
- **Scoliodon** is also known as **dogfish**. It is a cartilaginous fish.
- **Seahorse** (Hippocampus) is a bony fish.
- Most fish exchange gases like oxygen and carbon dioxide using gills that are protected under the gill covers on both sides of the pharynx (throat). Gills are tissues that are like short threads, protein structures called filaments.
- During winter ice forms on top of the water. Beneath the layer of ice there are layers of water where the temperature is more than 0°C. This is why the fishes live.
- **Amphibians** (Class-Amphibia) are any member of the group of **vertebrate** animals characterized by their ability to exploit both **aquatic** and **terrestrial** habitats. The name amphibian, derived from the ancient Greek term 'amphibios' which means 'both kinds of life'.
- **Reptiles** are tetrapod animals in the class **Reptilia**, comprising today's turtles, crocodiles, snakes, amphisbaenians, lizards and their extinct relatives.
- The **king cobra** (*Ophiophagus hannah*) is a venomous snake species in the family Elapidae, endemic to forests from India through South-East Asia. It is threatened by habitat destruction and has been listed **Vulnerable** on the IUCN Red List since 2010. It is the world longest venomous snake. Adult king cobras are about 3.18 to 4 metres long. The female king cobra lays eggs (20 to 50 eggs) in the nest (**Oviparous**). Parental care is found in this snake i.e. guards its own nest till the emerging of offsprings from the eggs.

Non-poisonous and poisonous snakes

- Most snakes are non-poisonous but some snakes as krait, cobra and viper are poisonous snake.
- Fangs are sharp, long, hollow or grooved teeth that are connected to a small sac in the snake's head behind its eyes. These sacs produce a poisonous liquid called venom. For some snakes with really long fangs, the fangs will fold back into the mouth so they do not bite themselves.
- Snake venom may contain twenty or more toxins.
- The cobra and krait venoms are neurotoxic and cardiotoxic while the venom of viper is vasculotoxic and haemotoxic.
- The poison gland of snakes are homologous to the salivary glands of vertebrates.

- Bats are **mammals** of the order **Chiroptera**, with their forelimbs adopted as wings, they are the only mammals naturally capable of true and sustained flight. Bats are more manoeuvrable than birds, flying with their very long spread-out digits covered with a thin membrane or patagium.
- The smallest bat (arguably the smallest extant mammal) is Kitti's hog-nosed bat (bumblebee bat), which is about 29-34 mm in length and 15 cm across the wings. The largest bats are the **flying foxes** (fruit bat) and the giant **golden-crowned flying fox**, which have a wingspan of about 1.7 metre.
- **The blue whale** (*Balaenoptera musculus*) is a marine mammal measuring upto 33 metre in length and with a maximum recorded weight of more than 180 metric tonnes. It is the largest animal known to have ever existed. Its lifespan is 80-90 years.
- **Sperm Whale** (*Physeter macrocephalus*) is the largest of the toothed whales measuring about 55 feet and weight about 35 to 45 tonnes. It is **carnivorous**.
- Sea Lion, Seal, Phoca, Walrus and Dolphins are aquatic mammals. **Dolphin** belongs to the order **Cetacea** while other belong to the order **Carnivora**.
- Dolphin is very intelligent animal.
- **River Ganga Dolphin** is the National Aquatic Animal of India. Its scientific name is *Platanista gangetica*.
- Sea cow is a giant herbivorous aquatic mammal.
- **Echidna** and **Platypus** are egg-laying mammals.
- **Nilgai** (*Boselaphus tragocamelus*) is the largest Asian **antelope** (It is not a cow). It is one of the most commonly seen wild animals of central and northern India, often seen in farmlands or scrub forests.
- **Apes** (super family-Hominoidea) are tailless primate of families **Hylobatidae** (gibbons-small ape) and **Hominidae** (chimpanzees, bonobos, orangutans, gorillas and human beings - great apes).
- Human beings are categorized zoologically as member of broader ape superfamily, they are usually placed within their own subcategories on account of their larger brain size, more advanced cognitive abilities (particularly the ability to speak) and striding two legged gait.
- The capacity of an adult human cranial cavity is 1200 - 1700 cm³.

Question Bank

1. Which of the following is the correct order of plant species from higher to lower order?
 - (a) Class – Phylum – Order – Plant Kingdom
 - (b) Order – Class – Phylum – Plant Kingdom
 - (c) Class – Plant Kingdom – Phylum – Order
 - (d) Plant Kingdom – Phylum – Class – Order

U.P.P.C.S. (Pre) 2024

Ans. (d)

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- **Order** : A group within a class, such as the order Rosales (within Magnoliopsida).
- **Family** : A group within an order, such as the rose family (Rosaceae).
- **Genus** : A group within a family, such as the genus Rosa (roses).
- **Species** : The most specific level, representing a group of plants that can interbreed and produce fertile offspring, such as Rosa rubiginosa (dog rose).

2. **A group of individual organism with fundamental similarities is called as a :**

- (a) Family (b) Species
(c) Genus (d) Order

Chhattisgarh P.C.S. (Pre) 2024

Ans. (b)

A group of individual organism with fundamental similarities is called as a species. When the group of related species have more common characters among them they are referred to as genus. A family is a broader group that contains multiple genera. Families which exhibits few similar characters are placed under the same order.

3. **Biologists have so far known, found and identified a large number of species in the plant and animal kingdom. In terms of numbers, the largest found and identified so far is from among the –**

- (a) Fungi (b) Plants
(c) Insects (d) Bacteria

I.A.S. (Pre) 1996

Ans. (c)

All insects belong to the group 'Insecta' which is a class of phylum Arthropoda. Class Insects represents the largest number of world species. Over 1.5 million living animal species have been described - of which more than 1 million are insects.

4. **Who discovered bacteria?**

- (a) Fleming (b) Lamble
(c) Temin (d) Leeuwenhoek

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

Antony Van Leeuwenhoek is regarded as the father of microbiology. He is known for the discovery of bacteria. He was born on October 24, 1632 in Delft, Netherlands.

5. **Bacteria can be seen by :**

- (a) Naked eye (b) Compound microscope
(c) Hand lens (d) None of the above

U.P.P.C.S. (Pre) 1995

Ans. (b)

Bacteria are microscopic living organisms. Most of the bacterias are just 0.5 to 5.0 micron (μ) in size but some of them are about 0.5 mm or more in size. Some of the microscopic organisms can be seen by a compound microscope as it can magnify an object up to 1500-2000 times.

6. **Assertion (A) : Some bacteria can synthesize their own food.**

Reason (R) : These bacteria have green matter called chloroplasts.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 2000

Ans. (c)

Bacteria have a wide range of diets. Some are heterotrophs and others are autotrophs. Most heterotrophic bacteria are saprobes (they absorb dead organic material like rotting flesh). Some of these parasitic bacteria kill their host while others help their host. Autotrophic bacteria make their own food, either by photosynthesis (which uses sunlight, carbon dioxide, and water to make food) or by chemosynthesis (which uses carbon dioxide, methane and chemicals like ammonia, hydrogen sulphide to make food). Phototrophic bacteria do not have chloroplast. Bacteriochlorophylls are photosynthetic pigments that occur in various phototrophic bacteria.

7. **Which of the following statement about bacteria is correct :**

- (a) They are on the borderline of living and non-living
(b) They are on the borderline of plants and animals

- (c) A is true but R is false.
(d) A is false but R is true.

U.P.P.C.S. (Pre) 1995

Ans. (a)

Amoeba is an unicellular organism which has the ability to alter its shape, primarily by extending and retracting pseudopods. Amoeba reproduces by fission. Binary fission is a type of asexual reproduction practiced by unicellular organisms in which the parent cell divides into two daughter cells. Generally unicellular organisms reproduce by asexual methods only while some unicellular eukaryotes (like Paramecium) are capable of both asexual and sexual reproduction.

- 11. In which organisms does spore formation occur as a mode of reproduction ?**

- (a) Fungi (b) Algae
(c) Yeast (d) Fern

70th B.P.S.C. (Pre) 2024

Ans. (*)

Spore formation is a common method of asexual reproduction in organisms like bacteria, fungi, algae, and some plants like mosses and ferns, where spores are produced and can develop into new individuals without fusion of reproductive cells. Hence, more than one option are correct.

- 12. Consider the following :**

1. Bacteria
2. Fungi
3. Virus

Which of the above can be cultured in artificial/synthetic medium?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2021

Ans. (a)

Bacteria and fungi can be cultured in artificial/synthetic medium. Many of them can be grown on nutritive media (e.g. nutrient agar, dextrose agar) to form colonies that can be seen with naked eyes. Viruses require a living host cell for replication. They cannot be cultured in artificial/synthetic medium due to lack of metabolic machinery of their own. Viruses can be grown *in vivo* (within a whole living organism, plant, or animal) or *in vitro* (outside a living organism in cells in an artificial environment). Infected host cells (eukaryotic or prokaryotic) can be cultured and grown, and then the growth medium can be harvested as a source of virus.

- (a) Both A and R are individually true, and R is the correct explanation of A.
- (b) Both A and R are individually true, but R is not the correct explanation of A.

13. Warm-blooded animals are those which are able to keep their body temperature –

- (a) Lower than environment
- (b) Higher than environment
- (c) Constant temp. all time
- (d) Equal to the atmospheric temp.

R.A.S./R.T.S. (Pre) 1996

Ans. (c)

Warm-blooded creatures like mammals and birds are able to keep their bodies at a constant temperature. They do this by generating their own heat when they are in a cooler environment and by cooling themselves in a hotter environment. This process of maintaining a consistent environment inside the body is called homeostasis.

14. Which one of the following statements is not correct?

- (a) All echinoderms are viviparous.
- (b) Roundworm has no circulatory system.
- (c) In bony fishes, swim bladder is usually present.
- (d) In cartilaginous fishes, fertilization is internal.

I.A.S. (Pre) 2002

Ans. (a)

Echinoderm is the common name given to any member of the Phylum Echinodermata of marine animals. The adults are recognizable by their (usually five-point) radial symmetry and include such well-known animals as starfish, sea urchins, sand dollars and sea cucumbers. All echinoderms are not viviparous. Fertilization is usually external in them and development is indirect with free-swimming larva. Hence, statement of option (a) is incorrect, while other three statements are correct.

15. Which one of the following creatures have no blood but respire ?

- (a) Hydra
- (b) Cockroach
- (c) Earthworm
- (d) Fish

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

Hydra has no blood but respire. In hydra both, the gaseous exchange and excretion occur by simple diffusion.

16. Of the following which one is not a fish?

- (a) Starfish
- (b) Sawfish
- (c) Pipefish
- (d) Guitarfish

Chhattisgarh P.C.S. (Pre) 2011

Ans. (a)

Starfish popularly called sea stars are related to the phylum Echinodermata while true fishes belong to the phylum Chordata. Starfish do not have gills, scales or fins which are basic characteristics of fishes.

17. Of the following, which one is a true fish ?

- (a) Silverfish
- (b) Crayfish
- (c) Jellyfish
- (d) Catfish

Uttarakhand P.C.S. (Mains) 2002

Ans. (d)

Silverfish are nocturnal insects. Their abdomens taper at the end, gives them a fish like appearance due to grayish blue and metallic shine as they get older.

Crayfish, also known as crawfish, crawdads or mudbugs, are fresh water crustaceans resembling small lobsters to which they are related taxonomically. They are the members of the superfamilies Astacoidea.

Jellyfish is also known as Aurelia. They are classified as free-swimming marine animals consisting of a gelatinous umbrella-shaped bell.

Catfish (order Siluriformes) are a diverse group of ray-finned fish. Named for their prominent barbels which resemble a cat's whiskers, catfish range in size and behaviour.

18. Which one of the following is a true fish?

- (a) Starfish
- (b) Jellyfish
- (c) Hagfish
- (d) Seahorse

R.A.S./R.T.S. (Pre) 1997

Ans. (d)

Starfish or Sea stars are star-shaped echinoderms belonging to the class Asteroidea. Jellyfish (Cnidaria) and Hagfish (Cyclostomata) are also not true fishes.

Seahorse is the name given to 45 species of small marine fish in the genus Hippocampus which is a bony fish.

19. Which of the following is fish ?

- (a) Sea cucumber
- (b) Sea cow
- (c) Seahorse
- (d) Sea lion

U.P. Lower Sub. (Pre) 2008

Ans. (c)

See the explanation of above question.

20. How many bones are in Shark Fish?

- (a) 100
- (b) 0
- (c) 200
- (d) 300

Chhattisgarh P.C.S. (Pre) 2011

Ans. (b)

Shark Fish contains no bones in its body. Sharks and other cartilaginous fish (skates and rays) have skeletons made of cartilage and connective tissues.

21. Which of the following is not a bird?

- (a) Golden Mahseer
- (b) Indian Nightjar

(c) Spoonbill

(d) White Ibis

I.A.S. (Pre) 2022

Ans. (a)

Among the given options, Golden Mahseer is not a bird. It is an endangered species of cyprinid fish that is found in rapid streams, riverine pools, and lakes in the Himalayan region. Its scientific name is *Tor putitora*. Indian Nightjar, Spoonbill and White Ibis belong to group aves (birds).

22. Which is the largest mammal?

- (a) Whale (b) African Elephant
(c) Hippopotamus (d) Polar bear

M.P.P.C.S. (Pre) 2010

Ans. (a)

Whales are the largest mammals ever known to have lived on earth. Whales range in size from the blue whale, the largest animal known to have ever existed at 33 m and more than 180 tons to pygmy species such as the pygmy sperm whale at 3.5 m (11 ft).

23. From which of following groups does the whale belongs –

- (a) Fishes (b) Mammals
(c) Reptiles (d) Amphibians

R.A.S./R.T.S. (Pre) 1993

Ans. (b)

See the explanation of above question.

24. Which one is a mammal?

- (a) Shark (b) Squid
(c) Octopus (d) Whale

R.A.S./R.T.S. (Pre) 2010

Ans. (d)

See the explanation of above question.

25. Which of the following species is the largest of the 'toothed-whales'?

- (a) Finback whale (b) Blue whale
(c) Sperm whale (d) Humpback whale

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (c)

The sperm whale (*Physeter macrocephalus*) is the largest of the toothed whales with males growing up to 20 metres in length. The mass of sperm whales ranges between 35 to 45 tons. Females only weigh about 1/3 of the males.

26. Which among the following one lays eggs and does not produce young ones directly?

- (a) Echidna (b) Kangaroo

(c) Porcupine

(d) Whale

I.A.S. (Pre) 2008

Ans. (a)

Echidnas, sometimes known as spiny anteaters, belong to the family Tachyglossidae in the monotreme order of egg-laying mammals. The four extant species, together with the Platypus are the only surviving members of that order and are the only living mammals that lay eggs. Their diet consists of ants and termites but they are not closely related with the true anteaters of America. They live in Australia and New Guinea.

27. Which of the following mammals lays eggs?

- (a) Loris (b) Kangaroo
(c) Hedgehog (d) Echidna

U.P. P.C.S. (Pre) 2022

Ans. (d)

See the explanation of above question.

28. Mammal capable in flying is –

- (a) Jaguar (b) Ostrich
(c) Pellican (d) Bat

M.P.P.C.S. (Pre) 2010

Ans. (d)

Bats are the mammals of the order Chiroptera whose forelimbs form webbed wings, making them the only mammals naturally capable of true and sustained flight. Bat does not belong to the group aves (birds).

29. Amongst the following flying organisms, which one does not belong to the group, Aves?

- (a) Bat (b) Crow
(c) Eagle (d) Parrot

U.P.P.C.S. (Mains) 2011

Ans. (a)

See the explanation of above question.

30. Consider the following statements :

Statement-I : The Indian Flying Fox is placed under the 'vermin' category in the Wild Life (Protection) Act, 1972.

Statement-II: The Indian Flying Fox feeds on the blood of other animals.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I

- (c) Statement-I is correct, but Statement-II is incorrect
(d) Statement-I is incorrect, but Statement-II is correct

I.A.S. (Pre) 2024

Ans. (*)

The nectar and fruit-eating large bat Indian Flying Fox (*Pteropus giganteus*) is generally considered a vermin as they raid orchards. It had a similar official status under erstwhile Schedule V of the Wildlife (Protection) Act of 1972 until it was put on the Schedule II list entailing a higher degree of protection (by the Amendment Act of 2022 w.e.f. 1-4-2023). As per amended Section 62 of Wildlife (Protection) Act, 1972, the Central Government may, by notification, declare any wild animal specified in Schedule II to be vermin for any area and for such period as may be specified therein and so long as such notification is in force, such wild animal shall be deemed not to be included in Schedule II for such area and for such period as specified in the notification. Schedule V of the Wildlife (Protection) Act of 1972 (which listed vermin category of animals) was omitted by the Amendment Act of 2022. Hence, at present, the Indian Flying Fox is not directly placed under the 'vermin' category in the Wild Life (Protection) Act, 1972. Hence, statement-I is incorrect in the present context. Indian Flying Fox is a frugivorous animal and feeds primarily on fruits, nectar, and flowers, not on the blood of other animals. It helps in seed dispersion and pollination, making them an integral part of the ecosystem. Hence, statement-II is also incorrect.

31. Consider the following animals –

1. Sea cow 2. Seahorse
3. Sea lion

Which of the above is/are mammal/mammals?

- (a) Only 1 (b) 1 and 3
(c) 2 and 3 (d) All of these

I.A.S. (Pre) 2013

Ans. (b)

Sea cow is a huge herbivorous sea mammal. Seahorse is a small marine bony fish. Sea lion is also a mammal. Hence option (b) is correct.

32. Dolphins are classified in –

- (a) Fish (b) Amphibians
(c) Reptile (d) Mammals

40th B.P.S.C. (Pre) 1995

Ans. (d)

Dolphins are highly intelligent aquatic mammals and are part of the family of toothed whales. They are found worldwide, mostly in shallow seas of the continental shelves and are carnivores, mostly eating fish and squid. India has declared Ganga Dolphin as the national aquatic animal.

33. From the following option, seal is the species of–

- (a) fish (b) bird
(c) reptiles (d) mammal

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (d)

Seals are semi-aquatic mammals, they are in a group called pinnipeds means fin-footed. All seals are carnivores. Their diet consists of mostly fish, crustaceans, and shellfish.

34. Amphibians are –

- (a) Very fastly moving boats
(b) Animals living in water only
(c) Animals living on land only
(d) Animals living equally in water and on land both

38th B.P.S.C. (Pre) 1992

Ans. (d)

The word 'amphibian' is derived from the ancient Greek term amphibios, which means 'both kinds of life'. The term was initially used as a general adjective for animals that could live on land or in water. They are ectotherms or cold-blooded animals, means they are unable to regulate their own body temperature.

35. Which organism among these breaths from skin ?

- (a) Fish (b) Pigeon
(c) Frog (d) Cockroach

Jharkhand P.C.S. (Pre) 2010

Ans. (c)

The frog is an amphibian creature and can live in water and land both. They are cold-blooded animal who often change their body temperature according to climate. The frog's skin is composed of thin membranous tissue that is quite permeable to water and contains a large network of blood vessels. The thin membranous skin allows the respiratory gases to readily diffuse directly down their gradients between the blood vessels and the surrounding. When the frog is out of the water, mucus gland in the skin keeps the frog moist, which helps absorb dissolved oxygen from the air. So a frog can breathe by skin and lungs both.

36. Frogs are cold-blooded animals –

- (a) True (b) False
(c) None of the above (d) All of the above

Chhattisgarh P.C.S. (Pre) 2011

Ans. (a)

Frogs are cold-blooded animals, it is true. Some other cold-blooded animals are snakes, lizards, fish, crocodile etc.

37. During the metamorphosis of frog –

- (a) Hind legs disappear
- (b) Ingestion of food is slowed down
- (c) Skin becomes rough
- (d) Intestine is shortened

U.P.P.S.C. (GIC) 2010

Ans. (d)

Metamorphosis is a biological process by which an animal physically develops after birth or hatching, involving a conspicuous and relatively abrupt change in the animal's body structure through cell growth and differentiation. Thyroid hormone controls remodelling of the tadpole intestine during the climax of amphibian metamorphosis. During the metamorphosis of frog, intestine shortens in length which helps to digest flesh food.

38. Gavialis crocodile are mostly found in ?

- (a) Ganga
- (b) Godavari
- (c) Krishna
- (d) Kaveri

R.A.S./R.T.S. (Pre) 1993

Ans. (a)

The Gharial, also known as Gavialis is a crocodilian of family Gavialidae mostly found in Ganga River. The large breeding population of gharial is found in Bangladesh, Nepal and India.

39. Which is an atrocious animal?

- (a) Penguin
- (b) Whale
- (c) Otter
- (d) Tortoise

39th B.P.S.C. (Pre) 1994

Ans. (c)

Among the given options, otter is an atrocious animal. Otters are said to be murderous, necrophilic aqua-weasels. Otters are mammals of the subfamily Lutrinae, one of two subfamilies contained within family Mustelidae. Otters are also called serial-killing fur monsters of the sea.

40. Consider the following animals :

1. Hedgehog
2. Marmot
3. Pangolin

To reduce the chance of being captured by predators, which of the above organisms rolls up/roll up and protects/protect its/their vulnerable parts?

- (a) 1 and 2
- (b) 2 only
- (c) 3 only
- (d) 1 and 3

I.A.S. (Pre) 2021

Ans. (d)

Hedgehogs are spiny mammals of the subfamily Erinaceinae, in the eulipotyphlan family Erinaceidae. All species of hedgehogs can roll into a tight ball in self-defense, causing all of the spines to point outwards. The hedgehog's back contains two large muscles that control the position of the quills. When the creature is rolled into a ball, the quills on the back protect the tucked face, feet, and belly, which are not quilled.

Marmots are large herbivores ground squirrels in the genus Marmota. They do not roll up when threatened. Marmots live in burrows and hide themselves in holes when there is a predator.

Pangolins, sometimes known as scaly anteaters are mammals of the order Pholidota. The name "pangolin" comes from the Malay word pengguling, meaning "one who rolls up". The physical appearance of a pangolin is marked by large, hardened, overlapping, plate-like scales, which are soft on newborn pangolins, but harden as the animal matures. The pangolin's scaled body is comparable in appearance to a pine cone. It can curl up into a ball when threatened, with its overlapping scales acting as armor, while it protects its face by tucking it under its tail. The scales are sharp, providing extra defense from predators.

41. Consider the following statements regarding the Indian squirrels :

1. They build nests by making burrows in the ground.
2. They store their food materials like nuts and seeds in the ground.
3. They are omnivorous.

How many of the above statements are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None of these

I.A.S. (Pre) 2023

Ans. (b)

Indian squirrels, like many squirrel species, are primarily arboreal creatures, which means they live in trees. They build nests, often called dreys, in trees, not burrows in the ground. Hence, statement 1 is incorrect.

Similar to many other squirrel species, Indian squirrels also hoard food. They store their food (like nuts and seeds) in various places, which can often include burying them in the ground, so they can retrieve them later when it turns to conditions of food scarcity like winter season. Hence, statement 2 is correct.

Indian squirrels are omnivores. They feed mainly on nuts and fruits but will also eat seeds, insects, small mammals and reptiles, eggs, and even sometimes chicks of birds. Hence, statement 3 is correct.

42. Which one of the following varieties of dogs is unable to bark ?

- (a) Dingo (b) Alsatian
(c) Dalmatian (d) Labrador

U.P.P.C.S. (Mains) 2010

Ans. (a)

Dingoes are commonly found throughout Australia and in scattered groups across south-east Asia. Its scientific name is *Canis lupus dingo*. Dingoes do not bark but howl like wolves.

43. The bear commonly performing on streets is –

- (a) Sloth bear (b) Sun bear
(c) Brown bear (d) Cat bear

U.P.P.C.S. (Mains) 2008

Ans. (a)

The bear which we see commonly performing on streets is sloth bear.

44. Which of the following is not one of the animals that has been taught to use language with some success?

- (a) Chimpanzee (b) Parrot
(c) Dog (d) Dolphin

U.P. P.C.S. (Mains) 2017

Ans. (c)

The Chimpanzee, Parrot and Dolphin can be trained to use language with some success but a dog can not be trained to use language.

45. Which of the following property separate a man/human from apes ?

- (a) Evince a desire to learn
(b) By a rudimentary sense of smell
(c) Contrary thumb
(d) Broader chin

R.A.S./R.T.S. (Pre) 1994

Ans. (a)

Human is the smartest creature in the animal kingdom. They are the only creature, who has a desire to learn and the ability to express.

46. Among the following, which one is not an ape?

- (a) Gibbon (b) Gorilla
(c) Langur (d) Orangutan

I.A.S. (Pre) 2008

Ans. (c)

Apes are human's closest living relatives. They do not have tails. In the animal kingdom, they belong to the group (Super- family) Hominoidea e.g. gibbon, gorilla, orangutans, chimpanzees and humans. Langur is the general name given to numerous species of Asian monkeys. They have long tails. They are not Apes. They belong to the subfamily Colobinae.

47. Smallest man-like ape is :

- (a) Gibbon (b) Chimpanzee
(c) Gorilla (d) Orangutan

U.P. P.C.S. (Pre) 2016

Ans. (a)

The gorilla, chimpanzee, and orangutan are called great apes in recognition of their comparatively large size and humanlike features. The gibbons are called lesser apes. They are smallest, cleanest and gentlest man like an ape.

48. Which one of the following makes a tool with a stick to scrape insects from a hole in a tree or a log of wood?

- (a) Fishing cat (b) Orangutan
(c) Otter (d) Sloth bear

I.A.S. (Pre) 2023

Ans. (b)

Orangutans are known for their high cognitive abilities and have been observed using tools in the wild. One of the most common tools that orangutans use is a stick, which they use for a variety of purposes. Orangutans use sticks to remove insects from tree holes, poke logs for honey, and when eating Neesia fruits, which are known for their irritant hairs. Once they use the sticks to remove these impediments, they can feast on the fruit's seeds unencumbered.

49. The Panda belongs to the same family as that of :

- (a) Bear (b) Cat
(c) Dog (d) Rabbit

I.A.S. (Pre) 2009

Ans. (a)

The panda is a member of the bear family, Ursidae.

50. Which one of the following animals have three chambered heart?

- (a) Monkey (b) Crocodile
(c) Fish (d) Frog

U.P. Lower Sub. (Pre) 2013

Ans. (d)

Amphibians and most reptiles have a heart with three chambers- two atria and a single ventricle. The heart of the frog has three chambers, one ventricle and two atria. Fish has two chambered heart while monkey and crocodile has four chambered heart.

51. Three chambered heart is found in :

- (a) Mammals (b) Birds
(c) Amphibians (d) Pisces

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (c)

See the explanation of above question.

52. Which out of the following does not have a gallbladder?

- (a) Camel (b) Giraffe
(c) Rat (d) Fish

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (*)

All members of the deer family except for the musk deer, all of the equine family (including zebras), camels, giraffes, elephants, rhinoceroses, whales, some birds (such as doves), rats and some fish do not have gallbladder.

53. In which of the following animals hooves are not found?

- (a) Deer (b) Jackal
(c) Zebra (d) Horse

U.P.P.C.S. (Pre) 2017

Ans. (b)

Among the given options except Jackal, all are hooves animals. Jackal is the species of canidae. The scientific name of common Jackal is *Canis aureus*. The feet of Jackals are padded with nails.

54. Nilgai belongs to the following family :

- (a) Cow (b) Goat
(c) Sheep (d) Deer

U.P.P.C.S. (Mains) 2006

Ans. (d)

Nilgai is the largest Asian antelope (deer-like) species. Antelope are creatures having strong permanent horns. There are more than 90 species of antelope.

55. Match List-I (Indian wild life species) with List-II (Scientific names) and select the correct answer using the codes given below :

List-I (Indian Wildlife Species)	List-II (Scientific Names)
A. Asiatic wild ass	1. <i>Boselaphus tragocamelus</i>
B. Barasingha	2. <i>Cervus duvaucelii</i>
C. Chinkara	3. <i>Equus hemionus</i>
D. Nilgai	4. <i>Gazella bennettii</i>

Code :

	A	B	C	D
(a)	2	3	1	4
(b)	3	2	4	1
(c)	2	3	4	1
(d)	3	2	1	4

I.A.S. (Pre) 2002

Ans. (b)

The correctly matched lists are as follows :

Asiatic wild ass	– <i>Equus hemionus</i>
Barasingha	– <i>Cervus duvaucelii</i>
Chinkara	– <i>Gazella bennettii</i>
Nilgai	– <i>Boselaphus tragocamelus</i>

56. What is the scientific name of Leopard?

- (a) *Panthera leo*
(b) *Panthera pardus*
(c) *Panthera tigris*
(d) *Panthera uncia*

Chhattisgarh P.C.S. (Pre) 2024

Ans. (b)

The scientific name of Leopard is '*Panthera pardus*'. It is one of the five largest living species of the cat (*Felidae*) family in the genus *Panthera*. Other four are : Lion (*Panthera leo*), Tiger (*Panthera tigris*), Snow Leopard (*Panthera uncia*) and Jaguar (*Panthera onca*).

57. Consider the following statements:

1. Lions do not have a particular breeding season.
2. Unlike most other big cats, cheetahs do not roar.
3. Unlike male lions, male leopards do not proclaim their territory by scent marking.

Which of the statements given above are correct ?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2024

Ans. (a)

Lions are polygamous, with the dominant male of the pride breeding with multiple females in the group. There is no specific breeding season for lions. In captivity lions often breed every year, but in the wild they usually breed no more than once in two years. Hence, statement 1 is correct. Unlike most other big cats, including lions, tigers, leopards, and jaguars, cheetahs don't roar. They growl when there is danger, and usually only chirp, purr and meow. Hence, statement 2 is correct. Like male lions, male leopards also proclaim their territory by scent marking. Both male and female leopards use urine to mark their territories, and often after urinating a male will then scrape the ground to transfer the scent of his urine onto his feet to be carried during territorial boundary patrols. Leopards are also often seen rubbing their faces and necks on vegetation as they are walking along their territory. By doing this, the Leopard is transferring secretions from a cheek gland, which then carries olfactory messages to other Leopards. Hence, statement 3 is incorrect.

58. Consider the following statements :

Statement-I : Marsupials are not naturally found in India.

Statement-II : Marsupials can thrive only in montane grasslands with no predators.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is incorrect
- (d) Statement-I is incorrect but Statement-II is correct

I.A.S. (Pre) 2023

Ans. (c)

Marsupials are a diverse group of mammals belonging to the infraclass Marsupialia. One of the defining features of marsupials is their unique reproductive strategy, where the young are born in a relatively undeveloped state and then nurtured within a pouch. Living marsupials encompass a wide range of species, including kangaroos, koalas, opossums, Tasmanian devils, wombats, wallabies, and bandicoots, among others. Each species exhibits distinct adaptations and behaviors suited to its particular ecological niche.

Marsupials are primarily found in Australasia, Wallacea, and the Americas. They are not naturally found in India. However, marsupials can thrive in various habitats, not only in montane grasslands with no predators. For example, some marsupials live in deserts, woodlands, and savannas. They can be found in dry scrublands, temperate deciduous forests and tropical rainforests. Hence, Statement-I is correct while Statement-II is incorrect.

59. Match List-I with List-II and choose the right answer from the given code -

List-I (Symptoms)	List-II (Creature)
A. Wingless insect	1. Kiwi
B. Flightless birds	2. Silverfish
C. Limbless Reptiles	3. Kurma
D. Lungless creature	4. Snakes
	5. Fish

Code :

	A	B	C	D
(a)	1	3	2	5
(b)	2	1	4	5
(c)	2	1	3	4
(d)	3	1	4	2

I.A.S. (Pre) 1993

Ans. (b)

The correctly matched lists are as follows :

Wingless insect	–	Silverfish
Flightless bird	–	Kiwi
Limbless reptile	–	Snake
Lungless creature	–	Fish

60. Kiwi is –

- (a) A species of vulture found in Amazon forest only
- (b) Flightless bird found only in New Zealand
- (c) A desert snake
- (d) Fastest flying Chiropteran mammal of Australia

R.A.S./R.T.S. (Pre) (Re. Exam) 2013

Ans. (b)

Kiwi are flightless birds native to New Zealand in the genus Apteryx and Family Apterygial.

61. Which of the following is of organic origin ?

- (a) Coral
- (b) Emerald
- (c) Ruby
- (d) Topaz

Uttarakhand P.C.S. (Pre) 2010

Ans. (a)

Corals are marine invertebrates in the class Anthozoa of phylum Cnidaria. They typically live in compact colonies of many identical individual polyps. The group includes the important reef builders that inhabit tropical oceans and secrete calcium carbonate to form a hard skeleton.

62. Octopus is :

- (a) an arthropod
- (b) an echinoderm
- (c) a hemichordate
- (d) a mollusc

I.A.S. (Pre) 2003

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (d)

An octopus is a cephalopod mollusc of the order octopoda, also known as devilfish. It has two eyes and four pairs of arms and like other cephalopods, it is bilaterally symmetric.

63. The largest invertebrate is –

- (a) Octopus
- (b) Squid
- (c) Coral
- (d) Jellyfish

U.P.P.C.S. (Mains) 2004

Ans. (b)

Invertebrates are the animals that do not have a backbone. The colossal squid is the largest invertebrate. It's maximum size is 12-14 metre.

64. Among the following organisms, which one does not belong to the class of other three?

- (a) Crab
- (b) Mite

(c) Scorpion

(d) Spider

I.A.S. (Pre) 2014

Ans. (a)

Class of crab – Malacostraca
Class of mite – Arachnida
Class of scorpion – Arachnida
Class of spider – Arachnida
Hence, crab does not belong to the class of other three.

65. White leg-horn is a variety of :

(a) Parrot

(b) Peacock

(c) Fowl

(d) Owl

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

White leg-horn is a variety of fowl.

66. Ticks and mites are actually :

(a) Arachnids

(b) Crustaceans

(c) Insects

(d) Myriapods

I.A.S. (Pre) 2002

Ans. (a)

Ticks and mites belongs to one of the most diverse groups of class, Arachnids. They have 4 pairs of legs, although some juveniles only have 3 pairs gaining the fourth pair with their first molt. There is no external segmentation of the abdomen, individuals appear as a single body mass.

67. Termites are also known as :

(a) Ants

(b) Red ants

(c) White ants

(d) Black ants

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

Termites are eusocial insects that are classified at the taxonomic rank of infraorder Isoptera. Termites are also known as white ants.

68. Glow-worm is :

(a) A mollusca

(b) An insect

(c) A worm

(d) A nematode

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

Glow-worm is a medium to large sized invertebrate luminous insect, that emits light. The light of the glow-worm larva is given off by small tubes ending around the glow-worms anus, as a product of excretion.

69. The presence of parasites within parasites is a phenomenon known as :

(a) Commensalism

(b) Hyperparasitism

(c) Endoparasitism

(d) Ectoparasitism

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

A hyperparasite is a parasite whose host is also a parasite. This form of parasitism is especially common among entomophagous parasites. The term is used loosely to refer also to parasitoids whose hosts are parasites or parasitoids. Hyperparasites are found mainly among the wasp-waisted Apocrita within the Hymenoptera, and in two other insect orders, the Diptera (true flies) and Coleoptera (beetles).

70. Female Anopheles can be distinguished from female Culex because it sits :

(a) at right angle to the surface of substratum

(b) at an angle with substratum

(c) parallel to the surface of substratum

(d) More than one of the above

(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

There are differences between female culex and female anopheles in their feeding pattern, size, eggs, and habitat. Female Anopheles body rests at an angle to the substratum while Culex body rests parallel to the substratum. Adult Anopheles can be identified by their typical resting position: males and females rest with their abdomens sticking up in the air rather than parallel to the surface on which they are resting. Anopheles mosquito makes angles its body at around 45 degrees from the surface.

71. Which one of the following is not an insect ?

(a) Bed bug

(b) Spider

(c) House fly

(d) Mosquito

U.P.P.C.S. (Spl.) (Pre) 2008

U.P. U.D.A./L.D.A. (Mains) 2010

Jharkhand P.C.S. (Pre) 2021

Ans. (b)

Spiders belong to a group of animals called 'Arachnids'. Arachnids are creatures with two body segments, eight legs and no wings or antennae. All spiders are predators and many eat each other. Rest three options are insects. They have six legs, while spider has eight legs. Thus spider is not an insect.

72. Spiders are different from insects because they have :

(a) Six legs

(b) Eight legs

(c) Ten legs

(d) Twelve legs

R.A.S./R.T.S. (Pre) 1996

Ans. (b)

See the explanation of above question.

73. Which is not insect amongst following?

- (a) Butterfly
- (b) Cockroach
- (c) Mosquito
- (d) Spider

Chhattisgarh P.C.S. (Pre) 2011

Ans. (d)

See the explanation of above question.

74. The silk produced by spiders is called :

- (a) Tussar silk
- (b) Gossamer silk
- (c) Munga silk
- (d) Ahimsa silk

U.P. R.O./A.R.O. (Pre) 2021

U.P. R.O./A.R.O. (Pre) 2017

Ans. (b)

Spider silk is protein fiber spun by spiders. It is also known as 'Gossamer silk'. Spider silk is an extremely strong material. It is one of the most miraculous natural substances on Earth. Most building materials are either strong or elastic, but spider silk is both. It has been often described as stronger than steel (which is not quite accurate, but close).

75. At which stage in its life cycle, does the silkworm yield the fibre of commerce?

- (a) Egg
- (b) Larva
- (c) Pupa
- (d) Imago

I.A.S. (Pre) 2000

Chhattisgarh P.C.S (Pre) 2005

Ans. (c)

The silkworm produces the silk of commercial importance in pupa stage of its life cycle. The egg is the first stage of a silkworm's life cycle. The larva is the vegetative stage where growth takes place. In pupa stage, as the silkworm prepares to pupate, it spins a protection cocoon. About the size and colour of a cotton ball, the cocoon is constructed from one continuous strand of silk perhaps 1.5 km long. The silk cocoon serves as protection for the pupa. Imago is the adult stage which completes the life cycle of a silkworm.

76. Silkworm grows on –

- (a) Reetha leaves
- (b) Ficus leaves
- (c) Peach leaves
- (d) Mulberry leaves

Uttarakhand Lower Sub. (Pre) 2010

Ans. (d)

The silkworm is the larva or caterpillar of the domesticated silkworm, *Bombyx mori*. It is an economically important insect, being a primary producer of silk. It grows on leaves of the mulberry tree.

77. Which one of the following is a secretion from the body of an insect ?

- (a) Pearl
- (b) Coral
- (c) Lac
- (d) Gum

U.P.P.C.S. (Pre) 1996

Ans. (c)

Lac is the scarlet resinous secretion of a number of species of lac insects, of which the most commonly cultivated species is *Kerria lacca*. Cultivation begins when a farmer gets a stick (brood lac) that contains eggs ready to hatch and ties it to the tree to be infested. Thousands of lac insects colonize the branches of the host trees and secrete the resinous pigment. It is used in the manufacture of varnishes, sealing wax, shellac etc.

78. Which of the following is not obtained from insects?

- (a) Silk
- (b) Honey
- (c) Lac
- (d) Pearl

U.P.P.C.S. (Pre) 2018

Ans. (d)

Among the given options silk, lac and honey are obtained from insects, while pearl is produced within the soft tissue of a living shelled mollusk or another animal such as fossil conulariids. The most valuable pearls occur spontaneously in the wild (natural pearls which are rare), while cultured or farmed pearls are obtained from bivalve mollusks, such as pearl oysters (*Pinctada*) and freshwater mussels.

79. Which of the following organisms has white blood?

- (a) Lizard
- (b) Cockroach
- (c) House bug
- (d) Mosquito

U.P.P.C.S. (Pre) 2019

Ans. (b)

Cockroaches have white or colourless blood because they lack haemoglobin in their blood. Cockroaches do not use hemoglobin to carry oxygen. In fact their bloodstream is not used to carry oxygen either. They use a system of tubes called tracheae to bring the oxygen and remove carbon dioxide from their tissues.

80. The animal without red blood cells is :

- (a) frog
- (b) earthworm
- (c) snake
- (d) peacock
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (b)

Earthworms lack red blood cells, yet they have a unique type of haemoglobin called erythrocrurin. The blood of earthworms is red due to the presence of this haemoglobin dissolved in the plasma.

81. How do most insects respire ?

- (a) Through skin (b) Through gills
(c) By lungs (d) By tracheal system

I.A.S. (Pre) 2007

Ans. (d)

Insects require oxygen to live and produce carbon dioxide as a waste product just as we do. They do not have lungs nor do they transport oxygen through their circulatory systems. Instead, insects use a series of tubes called a tracheal system to perform oxygen exchange throughout the body.

82. The organisms "Cicada, Frog hopper and Pond skater" are:

- (a) Birds (b) Fish
(c) Insects (d) Reptiles

I.A.S. (Pre) 2024

Ans. (c)

The organisms "Cicada, Frog hopper and Pond skater" are insects.

The cicadas are a superfamily, the Cicadoidea, of insects in the order Hemiptera (true bugs). Cicadas are more than 3,000 species of sound-producing insects. Cicadas are found in tropical and temperate areas worldwide and occur in deserts, grasslands, and forests. They are physically distinguished by their stout bodies, broad heads, clear-membraned wings, and large compound eyes.

The frog hoppers, or the superfamily Cercopoidea, are insects in the order Hemiptera. They are numerous species of small hopping insects worldwide in distribution. Adults are capable of jumping many times their height and length, giving the group their common name, but many species are best known for their plant-sucking nymphs which produce foam shelters, and are referred to as "spittlebugs".

The Gerridae are a family of insects in the order Hemiptera, commonly known as pond skaters, water striders, water skimmers or puddle flies etc. They are anatomically built to transfer their weight to be able to run on top of the water's surface. They can be often seen running or skating in groups over the surface of a pond or stream. With their short front legs they capture insects that fall onto the water's surface.

83. Consider the following statements :

1. Tapeworm is a hermaphrodite.
2. Roundworm has separate sexes.
3. Filaria is caused by a nematode.
4. Guinea worm is an annelid.

Which of these are correct ?

- (a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 2, 3 and 4

I.A.S. (Pre) 2001

Ans. (b)

According to the question, statement 1, 2 and 3 are correct while statement 4 is not correct because guinea worm is a parasite organism whose scientific name is *Dracunculus medinensis*. *Dracunculus medinensis* or guinea worm is a nematode that causes dracunculiasis, also known as guinea worm disease. The disease is caused by the female which is up to around 800 mm in length and among the longest nematodes infecting humans.

84. Which of the following are detritivores?

1. Earthworms
2. Jellyfish
3. Millipedes
4. Seahorses
5. Woodlice

Select the correct answer using the code given below.

- (a) 1, 2 and 4 only (b) 2, 3, 4 and 5 only
(c) 1, 3 and 5 only (d) 1, 2, 3, 4 and 5

I.A.S. (Pre) 2021

Ans. (c)

A detritivore is an organism that eats dead or decaying plants or animals as food. Detritivore is a combination of the word detritus and the suffix-vore. In other words detritivores are heterotrophs that obtain nutrients by consuming detritus. By doing so, all these detritivores contribute to decomposition and the nutrient cycles. Detritivores are usually arthropods and help in the process of remineralization. Typical detritivorous animals include millipedes, springtails, woodlice, dung flies, slugs, many terrestrial worms (e.g. earthworm), sea stars, sea cucumbers, fiddler crabs etc. Seahorses and jellyfish are not detritivores as they are generally secondary consumers. Hence, option (c) is the correct answer.

85. Which one of the following is a filter feeder?

- (a) Catfish (b) Octopus
(c) Oyster (d) Pelican

I.A.S. (Pre) 2021

Ans. (c)

Filter feeders are a sub-group of suspension feeding animals that feed by straining suspended matter and food particles from water, typically by passing the water over a specialized filtering structure. Filter feeders can play an important role in clarifying water, and are therefore considered ecosystem engineers. They are also important in bioaccumulation and, as a result, as indicator organisms. Some animals that use this method of feeding are clams, oysters, krill, sponges, baleen whales, and many fish (including some sharks). Some

birds, such as flamingos and certain species of duck, are also filter feeders. Oysters like other shellfish have unique filter-feeding ability, in the sense it takes in water through its gills, which then filters it for nutrients like suspended fragments of plankton, while automatically draining out the rest of the water.

86. Consider the following:

- | | |
|--------------------|---------------|
| 1. Carabid beetles | 2. Centipedes |
| 3. Flies | 4. Termites |
| 5. Wasps | |

Parasitoid species are found in how many of the above kind of organisms?

- | | |
|---------------|----------------|
| (a) Only two | (b) Only three |
| (c) Only four | (d) All five |

I.A.S. (Pre) 2024

Ans. (b)

A parasitoid is an animal, typically an insect, that lays its eggs on or inside a host, with the larvae feeding on the host and eventually killing it. There are two general categories of parasitoids: endoparasitoids, which hatch within the host from eggs or larvae laid there by an adult female, and then feed and develop inside the host; and ectoparasitoids, which are fastened to the outside of the host and feed through the host skin, sucking out body fluids.

Most parasitoids are either wasps or flies, although a few species of beetles, twisted wing insects, moths, and other insects have been identified as parasitoids. Several genera of carabid beetles are ectoparasitoids as larvae.

Centipedes are predatory arthropods. Centipedes are predominantly generalist carnivorous, hunting for a variety of prey items that can be overpowered. Termites are primarily detritivores (also known as detritus feeders), needing water, cellulose, and shelter to survive. Termites consume a wide variety of decaying plant material, generally in the form of wood, leaf litter, and soil humus.

Hence, among the given organisms, parasitoid species are found in only three organisms.

87. Which of the following group of animals are nocturnal animals?

- | | |
|------------------------------|------------------------|
| (a) Housefly, Bedbug, Parrot | (b) Mosquito, Bat, Owl |
| (c) Mosquito, Sparrow, Deer | (d) Owl, Bats, Dog |

Uttarakhand U.D.A./L.D.A. (Pre) 2005

Ans. (b)

Nocturnal animals are more active at night than during the day. These animals sleep during the day, often in a burrow or den. Nocturnal creatures generally have highly developed senses of hearing, smell, and specially adapted eyesight. Mosquito, bat, owl and kiwi are some example of nocturnal animals.

88. Consider the following fauna :

- | | |
|------------------------|------------------|
| 1. Lion-tailed Macaque | 2. Malabar Civet |
| 3. Sambar Deer | |

How many of the above are generally nocturnal or most active after sunset?

- | | |
|---------------|--------------|
| (a) Only one | (b) Only two |
| (c) All three | (d) None |

I.A.S. (Pre) 2023

Ans. (a)

The lion-tailed macaque (*Macaca silenus*), also known as the wanderoo, is an Old World monkey endemic to the Western Ghats of India. The lion-tailed macaque is a rainforest dweller, often being found in the upper canopy of tropical moist evergreen forests or monsoon forests. It is diurnal, meaning it is active exclusively in daylight hours. They sleep at night in trees (typically, high in the canopy of rainforest). Hence, 1 is incorrect.

The Malabar large-spotted civet (*Viverra civettina*), also known as the Malabar civet, is a viverrid endemic to the Western Ghats of India. The Malabar civet is considered nocturnal and so elusive that little is known about its biology and ecology apart from habitat use. Hence, 2 is correct.

The sambar (*Rusa unicolor*) is a large deer native to the Indian subcontinent, South China and Southeast Asia. Sambars are in general mostly crepuscular (most active at dawn and dusk) animals, with significant nocturnal activity as well. Thus, 3 is not completely true and could be taken as incorrect.

89. Which of the following has the largest brain in proportion to its body size?

- | | |
|-------------|-----------------|
| (a) Ant | (b) Elephant |
| (c) Dolphin | (d) Human Being |

U.P.P.C.S. (Pre) 2017

Ans. (a)

The largest brain belongs to the sperm whale : about 7 kg (17.5 pounds). The blue whale, the largest animal on the planet, being twice longer and thrice heavier, has a brain weighing about 5 kg (12.5 pounds).

Human brain has an average weight of about 2.7 pounds (1.2 kg), variations between 1.1 and 1.4 kg being considered normal. Our brain represents about 1.5- 2% of our weight. The largest brain of terrestrial animal is that of elephant : about 10.5 pounds (4.78 kg). Still, the brain of the elephant makes less than 0.1% of its body weight.

Ant has the largest brain in proportion to its body size.

90. When ants bite, they inject :

- | | |
|-------------------------|------------------|
| (a) Glacial acetic acid | (b) Methanol |
| (c) Formic acid | (d) Stearic acid |

I.A.S. (Pre) 1999

Ans. (c)

The formic acid is produced by the ants. Chemically it is a simple carboxylic acid. When ants bite, they inject this acid.

91. Certain species of which one of the following organisms are well known as cultivators of fungi?

- (a) Ant (b) Cockroach
(c) Crab (d) Spider

I.A.S. (Pre) 2022

Ans. (a)

Many species of ants are well known as cultivators of fungi. The ant-fungus mutualism is a symbiosis seen between certain ants (tribe Attini) and fungal species, in which ants actively cultivate fungus much like humans farm crops as a food source. The leafcutter ant is a well-known example of this symbiosis. They are known to farm fungus in the Americas and the Caribbean. However, ants are not the only ground-dwelling arthropods which have developed symbioses with fungi. A mutualism with fungi is also noted in some species of termites in Africa.

92. Consider the following statements about mosquitoes :

1. Only males suck blood
2. Only females suck blood
3. Females suck plant juices
4. Males have bigger wings than females
5. Females have bigger wings than males

Of these :

- (a) Only 2 and 4 are correct.
(b) Only 1 and 2 are correct.
(c) Only 3 and 5 are correct.
(d) Only 2 and 5 are correct.

U.P. R.O./A.R.O. (Pre) 2016

Ans. (d)

Only female mosquitoes suck the blood of warm-blooded vertebrate birds and mammals and live on them as an ectoparasite. The male mosquitoes do not suck blood, they survive by feeding on flowers and fruit juices. Occasionally, females may also suck juices. Generally, female mosquitoes are larger than male mosquitoes. That is why option (d) is the correct answer.

93. Which one of the following animals stores water in the intestine?

- (a) Moloch (b) Camel
(c) Zebra (d) Uromastix

I.A.S. (Pre) 1994

Ans. (b)

Camels have some special abilities which enable it to walk for a long distance over sandy desert hills when there is neither water nor vegetation. That is why it is called 'The Ship of the Desert'. They have broad, flat, leathery pads with two toes on each foot. Their breathing rate is very slow. A camel can drink more than 100 litres of water at once and they store it in their intestine for future needs.

94. Which of the following animals can live for the longest duration without drinking water?

- (a) Giraffe (b) Camel
(c) Kangaroo (d) Kangaroo rat

U.P.P.C.S. (Pre) 2018

Ans. (d)

Among the given animals, Kangaroo rat can live for the longest duration without drinking water. The Kangaroo rat is a rodent native to the desert areas of North America. The Kangaroo rat adapted to live in the desert climate because it produces the most concentrated urine in all mammals. Their kidneys concentrate the urine so much that it becomes almost solid.

95. Which among the following animals excretes solid urine?

- (a) Kangaroo rat (b) Giraffe
(c) Squirrel (d) Crow

U.P. P.C.S. (Mains) 2017

Ans. (a)

See the explanation of above question.

96. In which one of the following kinds of organisms is the phenomenon found wherein the female kills the male after copulation?

- (a) Dragonfly (b) Honeybee
(c) Spider (d) Pit Viper

I.A.S. (Pre) 2008

Ans. (c)

Spider cannibalism is the act of a spider consuming all or part of another individual of the same species as food. In the majority of cases, a female spider kills and eats a male spider before, during or after copulation. But the strong male spider escapes and again copulate.

97. King Cobra is the only snake that makes its own nest. Why does it make its nest?

- (a) It is a snake-eater and the nest helps attract other snakes
(b) It is a viviparous snake and needs a nest to give birth to its offspring

- (c) It is an oviparous snake and lays its eggs in the nest and guards the nest until they are hatched
(d) It is a large, cold blooded animal and needs a nest to hibernate in the cold season

I.A.S. (Pre) 2010

Ans. (c)

The king cobra is unusual among snakes in such a way that the female king cobra is a very dedicated parent. She makes a nest for her eggs, scraping up leaves and other debris into a mound in which to deposit them and remains in the nest until the young hatch. A female usually deposits 20 to 40 eggs into the mound, which acts as an incubator.

98. The only snake that builds a nest is :

- (a) Chain Viper (b) King Cobra
(c) Krait (d) Saw-scaled viper

I.A.S. (Pre) 1995

Ans. (b)

See the explanation of above question.

99. The venom of cobra is–

- (a) Neurotoxic (b) Haemotoxic
(c) Both (a) & (b) (d) None of the above

Uttarakhand P.C.S. (Pre) 2010

Ans. (a)

The venom of the cobra is mainly neurotoxic which affects nervous tissues. It has some other major compound as cardiotoxic. Toxins presents are mainly protein and polypeptide.

100. For which one of the following snakes is the diet mainly composed of other snakes?

- (a) Krait (b) Russell's viper
(c) Rattlesnake (d) King Cobra

I.A.S. (Pre) 2008

Ans. (d)

The king cobra's diet consists primarily of other snakes, including rat snakes, small pythons and even other venomous snakes such as various members of the true cobras and the krait. When food is scarce, they may also feed on other small vertebrates, such as lizards, birds, and rodents.

101. Which snake of the following is nonpoisonous?

- (a) Cobra (b) Viper
(c) Coral-Snake (d) Python

M.P.P.C.S. (Pre) 2012

Ans. (d)

The Pythonidae, commonly known as pythons are a family of nonvenomous snakes found in Africa, Asia and Australia. Among its members are some of the largest snakes in the world.

102. Poison glands of snakes are homologous to –

- (a) Electric organs of fishes
(b) Stings of rays
(c) Sebaceous glands of mammals
(d) Salivary glands of vertebrates

I.A.S.(Pre) 1993

Ans. (d)

Poison glands of snakes are homologous to salivary glands of vertebrates. Snake venom is actually the refined form of its saliva.

103. Poisonous fangs of a snake are modified form of :

- (a) Mandibular teeth (b) Palatine teeth
(c) Canine teeth (d) Maxillary teeth

U.P. R.O./A.R.O. (Pre) 2016

Ans. (d)

Poisonous fangs of a snake are modified form of maxillary teeth. There are 1-3 pairs (usually 1 pair) of long, curved, sharp and pointed maxillary teeth attached to the poison glands of the upper jaw of a snake.

104. Consider the following:

- 1. Butterflies 2. Fish 3. Frogs**

How many of the above have poisonous species among them?

- (a) Only one (b) Only two
(c) All three (d) None

I.A.S. (Pre) 2024

Ans. (c)

A number of butterflies actually have evolved to be poisonous. Butterflies typically get the poison from the plants they eat as caterpillars - which are sometimes toxic. They then accumulate these toxins and they remain in the adult butterfly's body. Examples of poisonous butterflies include the African Giant Swallowtail, Atala Butterfly, and Monarch.

A number of fish species are poisonous. Poisonous fish contain toxins which are not destroyed by the digestive systems of animals that eat the fish. Species of puffer fish (the family Tetraodontidae) are the most poisonous fish in the world, and the second most poisonous vertebrate after the golden dart frog. Many fish species are venomous too. Reef Stonefish is the most venomous known fish in the world.

Many species of frog are known to be poisonous. Poisonous frogs produce and store alkaloid poisons or toxins in their skin, which makes them harmful to touch. They are commonly called poison arrow frogs or poison dart frogs. But although incredibly rare, there are some frog species that are venomous too.

Hence, option (c) is the correct answer.

105. Consider the following statements :

1. Some species of turtles are herbivores.
2. Some species of fish are herbivores.
3. Some species of marine mammals are herbivores.
4. Some species of snakes are viviparous.

Which of the statements given above are correct?

- (a) 1 and 3 only (b) 2, 3 and 4 only
(c) 2 and 4 only (d) 1, 2, 3 and 4

I.A.S. (Pre) 2019

Ans. (d)

When green sea turtle (*Chelonia mydas*) becomes adult it becomes strictly herbivores. Many species of fish are herbivores, like parrot fish which eats algae. Some marine mammals like dugong are herbivores. Some species of snakes like green anaconda are viviparous means they give birth live young ones rather than laying eggs like other reptiles. Hence, option (d) is the correct answer.

106. Dogs are herbivores –

- (a) True (b) False
(c) None of the above (d) All of the above

Chhattisgarh P.C.S. (Pre) 2011

Ans. (b)

The dog is not herbivorous animal. The assumption that dogs are natural omnivores remains to be proven, whereas the truth about dogs being natural carnivores is very well-supported by the evidence available to us.

107. The largest flower in the world is–

- (a) Lotus (b) Sunflower
(c) Rafflesia (d) Glory Lily

U.P.P.C.S. (Pre) 1998

39th B.P.S.C. (Pre) 1994

Ans. (c)

Rafflesia plant with no leaves, no roots, no stem is the biggest flower in the world. It sounds like the stuff of comic books or science fiction. 'It is perhaps the largest and most magnificent flower in the world' was how Sir Stamford Raffles described his discovery in 1818 of *Rafflesia Arnoldii*, modestly named after himself and his companion, surgeon-naturalist Dr. James Arnold.

108. Insectivorous plants grow in soil deficient in :

- (a) water (b) magnesium
(c) nitrogen (d) calcium
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (c)

Insectivorous plants grow in nitrogen-deficient soil and fulfill their nitrogen requirement by trapping small insects. Insectivorous plants are especially adapted for capturing and digesting insects and other animals by means of ingenious pitfalls and traps.

109. Which one of the following is an insectivorous plant?

- (a) Passion flower plant (b) Pitcher plant
(c) Night queen (d) Flame of the forest

I.A.S. (Pre) 2008

Ans. (b)

A pitcher plant is a carnivorous plant or an insectivorous plant. Carnivorous plants are plants that eat insects and other small animals. Carnivorous plants grow in a soil that has less nitrogen. All living things must have nitrogen. Carnivorous plants get nitrogen from the insects they eat.

110. Which one of the following parts of the pitcher plants becomes modified into a pitcher?

- (a) Stem (b) Leaf
(c) Stipule (d) Petiole

I.A.S. (Pre) 2007

Ans. (b)

The pitcher plant is a carnivorous plant with highly modified leaves in the form of pitchers that acts as pitfall traps for prey. The narrow pitcher leaves are tapered tubes that rise up to about 75 cm from the ground, with a mouth about 6 to 10 cm in circumference.

111. Which one of the following is not correctly matched?

- | | |
|---------------|------------------------|
| (a) Yeast | – Fermentation |
| (b) Casein | – Milk protein |
| (c) Aloe Vera | – Angiosperm |
| (d) Nepenthes | – Parasitic angiosperm |

U.P. R.O./A.R.O. (Pre) 2016

Ans. (d)

Pitcher plants or *Nepenthes* are able to synthesize their food by the process of photosynthesis, hence they are autotrophs. Usually, they grow on soils deficient in nitrogenous compounds and fulfill their required nitrogen by digesting insects. Other options are correctly matched.

112. Which of the following is an insectivorous plant?

- (a) *Nyctanthes* (b) *Nelumbo*
(c) *Nicotiana* (d) *Nepenthes*

U.P. P.C.S. (Pre) 2023

Ans. (d)

Nepenthes are insectivorous pitcher plant. Nepenthes species are perennial herbaceous plants and frequently grow in very acidic soil, though some are epiphytes. All the species are carnivorous plants that capture insects and other prey by means of modified pitcher-shaped leaves that function as passive pitfall traps.

113. Nepenthes khasiana a rare and endangered plant is found in –

- (a) Himachal Pradesh (b) Madhya Pradesh
(c) Meghalaya (d) Uttar Pradesh

U.P.P.C.S. (Mains) 2011

Ans. (c)

Nepenthes khasiana is India's only pitcher plant and is named after the Khasi Hill region of Meghalaya. They are mostly found in Garo, Khasi and Jaintia hills region of Meghalaya.

114. Turmeric is obtained from which part of the plant :

- (a) Root (b) Stem
(c) Fruit (d) Flower bud

U.P.P.C.S. (Pre) 2006

R.A.S./R.T.S.(Pre) 1999

Ans. (b)

Turmeric is obtained from the modified stem (rhizome) of the plant. It is a member of the curcuma botanical group, which is part of a ginger family of herbs, the Zingiberaceae. Its botanical name is Curcuma longa. All curcumas are perennial plants native to southern Asia. They grow in warm humid climates and thrive only in temperatures above 60° F (29.8°C). India, Srilanka, Fiji and Queensland (Australia) all have the climate, which is conducive to grow turmeric.

115. Turmeric powder is obtained from which part of turmeric plant ?

- (a) Dried fruit (b) Dried root
(c) Seed (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

See the explanation of above question.

116. Which of the following is the eatable part of turmeric plant ?

- (a) Root (b) Rhizome
(c) Fruit (d) Flower

M.P.P.C.S. (Pre) 1993

Ans. (b)

See the explanation of above question.

117. Turmeric is an example of underground stem, known as :

- (a) Bulb (b) Tuber
(c) Corm (d) Rhizome

U.P. R.O./A.R.O. (Mains) 2021

Ans. (d)

See the explanation of above question.

118. Which of the following underground produced vegetables is a modified root ?

- (a) Potato (b) Sweet Potato
(c) Bandda (d) Jimikand

U.P.P.C.S. (Mains) 2005

Ans. (b)

The sweet potato (Ipomoea batatas) is a dicotyledonous plant that belongs to the family Convolvulaceae. Sweet potato is a modified root cultivated by vegetative propagation.

119. The storage organ of which of the following plants is not a stem ?

- (a) Sugarcane (b) Ginger
(c) Potato (d) Sweet potato

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (d)

Sugarcane, potato, and ginger are modified stems i.e., their storage organ is stem, while the storage organ of sweet potato is root. It is developed by radicle.

120. Which of the following is a stem?

- (a) Turnip (b) Ginger
(c) Carrot (d) Sweet potato

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

Among the given options turnip, carrot and sweet potato are modified roots while ginger is a modified stem.

121. Ginger is a stem and not root because :

- (a) it stores food material
(b) it has nodes and internodes
(c) it grows horizontally in the soil
(d) it lacks chlorophyll
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

Stems get modified into underground structures for storage of food as seen in potato (tuber), ginger (rhizome), garlic (bulb), yam (corm). Presence of an eye (node) in potato, distinct nodes with internodes and scaly leaves in ginger/yam, a cluster of roots at the base of the reduced stem in garlic/onion, all indicate that these underground plant parts are modified stems.

122. Match the following :

- | | |
|-------------------|--------------------------|
| A. Rhizome | 1. Garlic |
| B. Tuber | 2. Jimikand-suram |
| C. Bulb | 3. Ginger |
| D. Corm | 4. Potato |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 4 | 2 | 1 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 3 | 1 | 2 | 4 |
| (e) | 3 | 2 | 1 | 4 |

Chhattisgarh P.C.S (Pre) 2013

Ans. (c)

Tubers, rhizomes, corms and bulbs actually all serve the same purpose, but in a different way.

Plants that have rhizomes include ginger, turmeric, sugarcane etc.

Common examples of plants with tubers include potatoes, cassava and dahlias.

Examples of bulbs include onion, garlic, tulip, amaryllis, narcissus.

Examples of corm include Jimikand, crocus, colocasia, etc.

123. Which one of the following is a modified stem ?

- | | |
|-------------|------------------|
| (a) Carrot | (b) Sweet potato |
| (c) Coconut | (d) Potato |

I.A.S. (Pre) 1996

Ans. (d)

Potato is a starchy red or brown skinned modified stem called a tuber. Tuber is storage area for nutrient reserves of plants such as starch or sugar. The scientific name of potato plant is *Solanum tuberosum* and it is a member of the nightshade family of plants, Solanaceae.

124. Potato is a :

- | | |
|-----------|-----------|
| (a) Root | (b) Fruit |
| (c) Tuber | (d) Bulb |

M.P. P.C.S. (Pre) 2016

Ans. (c)

See the explanation of above question.

125. Pea plant is :

- | | |
|----------|-------------------|
| (a) Herb | (b) Flower |
| (c) Bush | (d) None of these |

39th B.P.S.C. (Pre) 1994

Ans. (a)

Pea plant is a dicotyledonous herbaceous annual plant. The pea plant may adapt to a variety of growing conditions, but it prefers cool damp weather with temperature ranging from 60 to 75 degree Fahrenheit. It belongs to a large family of plants, the fabaceae or leguminosae.

126. Consider the following plants:

- | | |
|---------------------|----------------------|
| 1. Groundnut | 2. Horse-gram |
| 3. Soybean | |

How many of the above belong to the pea family?

- | | |
|---------------|--------------|
| (a) Only one | (b) Only two |
| (c) All three | (d) None |

I.A.S. (Pre) 2024

Ans. (c)

Fabaceae (pea family) is a large family of angiosperms. It is also known as Leguminosae and the plants are commonly known as legumes. It is widely distributed all over the world.

The groundnut (*Arachis hypogaea*), also known as peanut, is a legume crop grown mainly for its edible seeds. It belongs to the family Fabaceae or Leguminosae, commonly known as the legume, bean, or pea family.

Horse-gram or kulthi (*Macrotyloma uniflorum*) is a legume native to tropical southern Asia, known for its distinct taste and texture. It also belongs to the family Fabaceae or pea family.

The soybean (*Glycine max*) is a species of legume native to East Asia, widely grown for its edible bean, which has numerous uses. It also belongs to the family Fabaceae or pea family.

127. Clove is a –

- | | |
|------------------|-------------------|
| (a) Buds of stem | (b) Bales of root |
| (c) Closed buds | (d) Seed |

U.P. Lower Sub. (Spl.) (Pre) 2002

U.P. Lower Sub. (Spl.) (Pre) 2003

Ans. (c)

Cloves are the aromatic closed, dried floral buds, which is used in Ayurvedic medicines, making food and as an anodyne for dental emergencies. The botanical name of clove tree is *Syzygium aromaticum* (*Eugenia caryophyllata*) which belongs to Myrtaceae family. Cloves are native of Indonesia and used as a spice in cuisines all over the world. The clove tree grows to a height ranging from generally 8-12 metres, having large oval leaves.

128. A clove represents to which of the following?

- | | |
|------------------|-------------------|
| (a) Terminal bud | (b) Accessory bud |
|------------------|-------------------|

- (c) Flower bud (d) Vegetative bud
U.P. P.C.S. (Pre) 2021

Ans. (c)

See the explanation of above question.

129. Which one of the following spices is a flower bud ?

- (a) Cumin (b) Clove
(c) Pepper (d) Turmeric
U.P.P.C.S. (Mains) 2011

Ans. (b)

See the explanation of above question.

130. 'Cloves' are –

- (a) Nodules of stem (b) Nodules of roots
(c) Leaves (d) Dried flowers
U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (d)

See the explanation of above question.

131. Cloves are obtained from which of the following parts of the plant?

- (a) Dried leaves (b) Dried stems
(c) Dried seeds (d) Dried flower buds
56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

See the explanation of above question.

132. Clove, a spice, is obtained from which part of the plant?

- (a) Fruit (b) Stem
(c) Root (d) Flower bud
(e) None of the above / More than one of the above
66th B.P.S.C. (Pre) 2020

Ans. (d)

See the explanation of above question.

133. Which of the following is not correctly matched–

- (a) Ginger – Rhizome
(b) Saffron – Seed
(c) Opium – Opium capsule
(d) Jute – Stem
U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (b)

Saffron is a spice derived from the flower of *Crocus sativus*, commonly known as the 'Saffron Crocus'. Saffron is obtained from the styles and stigmas of the flower of *Crocus sativus*. Other options are correctly matched.

134. Saffron is dried mixture of :

- (a) Leaf and stem
(b) Petals and roots
(c) Seed forming parts of flower
(d) Seeds and buds

U.P.P.C.S. (Pre) 2002

Ans. (c)

Saffron is obtained from the gynoecium parts (stigma and styles) of the flower of *Crocus sativus* which are seed forming parts of flower.

135. In the following pairs of plants and their edible parts, which pair is not correctly matched ?

- (a) Ginger – Rhizome
(b) Onion – Fleshy leaves
(c) Potato – Roots
(d) Coconut – Endosperm

U.P.P.C.S. (Mains) 2011

Ans. (c)

The edible part of the potato is tuber (modified stem), not root. Thus, option (c) is not correctly matched. Other options are correctly matched.

136. Which of the following parts of potato is edible?

- (a) Stem (b) Seed
(c) Root (d) Fruit

U.P.R.O./A.R.O. (Mains) 2013

Ans. (a)

See the explanation of above question.

137. Which one of the following is a fruit –

- (a) Potato (b) Radish
(c) Ladies finger (d) Sweet Potato

R.A.S./R.T.S. (Pre) 1999

Ans. (c)

Ladies finger (okra) is a flowering plant in the mallow family. Its botanical name is *Abelmoschus esculentus*. Its fruit is a capsule up to 18cm long with the pentagonal cross section, containing numerous seeds. The entire part of the fruit is edible. In remaining options, the edible part of the potato is tuber and root is the edible part of radish and sweet potato.

138. Among the following, which is not a true fruit?

- (a) Apple (b) Grape
(c) Date (d) Plum
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

The true fruit can be defined as the fruit, which is formed from the fertilized ovary of the flower. Mango, grapes, cherry, watermelon, lemon, date, plum, etc. are examples of true fruits. The false fruit can be defined as the fruit, which is formed from the ripened ovary along with some other parts of the flower. Some examples of false fruit are cashew-nut, which develops from the peduncle; apple, pear, gourd and cucumber, which develop from the thalamus; jack fruit and pineapple, which develop from the entire inflorescence, etc.

139. The type of fruit in which lychee can be placed in –

- (a) Drupe (b) Hesperidium
(c) Nut (d) One seeded berry

U.P.P.C.S. (Mains) 2010

Ans. (a)

Lychee is a juicy fruit belonging to the soapberry family. The fruit grows on an evergreen tree, with pinnate leaves. It is a drupe (like a plum, a cherry, or a mango), externally covered by a pink-red rough textured rind, easily removable.

140. The characteristic odour of garlic is due to –

- (a) a chloro compound
(b) a sulphur compound
(c) a fluorine compound
(d) acetic acid

I.A.S. (Pre) 1997

Chhattisgarh P.C.S. (Pre) 2005

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

Garlic is made up of sulphuric compounds (Allyl methyl sulphide) that render the pungent smell to it. It is also used as diaphoretic, diuretic, expectorant and for treating blood pressure and heart disease.

141. Peeling of onions causes tears as onions release –

- (a) Sulphonic acid (b) Sulfenic acid
(c) Amino acid (d) Carbolic acid

U.P.P.C.S. (Pre) 2015

Ans. (b)

Sulfenic acid found in onions irritates lacrymal glands and produces tears in eyes.

142. Pungency in chillies is due to the presence of :

- (a) Lycopene (b) Capsaicin
(c) Carotene (d) Anthocyanin

U.P. P.C.S. (Pre) 2016

Ans. (b)

Chillies are valued principally for their high pungency and colour. Pungency in chillies is due to the active constituent 'capsaicin', an alkaloid which is extracted from chillies and is also used in the medicine. The molecular formula of capsaicin is $C_{18}H_{27}NO_3$.

143. Red chillies are hot because these contain :

- (a) Ricin (b) Capsaicin
(c) Arecoline (d) Caffeol

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

Red chillies are hot because they contain capsaicin. Its chemical formula is $C_{18}H_{27}NO_3$.

144. The cause of sweet taste in fruits is :

- (a) Maltose (b) Ribose
(c) Lactose (d) Fructose

U.P.P.C.S. (Pre) 2017

Ans. (d)

Fructose is also known as fruit sugar. This type of sugar is mainly found in fruits. Fructose is a monosaccharide.

145. Which one of the following groups of organisms has significance in diagnosing the death by drowning?

- (a) Lichens (b) Protozoa
(c) Cyanobacteria (d) Diatoms

R.A.S./R.T.S.(Pre) 2008

Ans. (d)

Diatoms are useful in forensic studies. It has significance in diagnosing the death by drowning. Diatoms are photosynthetic algae and are found in almost every aquatic environment including fresh and marine waters, soils and almost at every humid place.

146. The mulberry fruit is :

- (a) Sorosis (b) Syconus
(c) Samara (d) Nut

U.P.P.C.S. (Mains) 2006

Ans. (a)

The mulberry fruit is also known as the sorosis. Sorosis is a multiple fleshy fruit that is derived from the ovaries of multiple flowers. The seeds are achenes, on the outside of a fleshy fruit. Pineapple and Ficus are also the examples of Sorosis fruit.

147. Which one of the following fibre is not a product of plant stem :

- (a) Sunn (b) Hemp

(c) Jute

(d) Cotton

U.P.P.C.S. (Pre) 2002

Ans. (d)

The fibres of sunn, hemp, and jute are obtained from stems, while cotton is obtained from the fruit of the cotton plant, called cotton balls. Cotton fibres are classified into two types namely lint and fuzz respectively. Lint fibre is more important for the commercial purpose than fuzz fibre.

148. Which among the following is a man-made cereal?

(a) Secale cereal

(b) Pomato

(c) Triticale

(d) None of the Above

U.P. P.C.S. (Mains) 2017

Ans. (c)

Triticale is the first man-made cereal crop. It has been obtained by crossing wheat (Triticum) with rye (Secale) in laboratories during the late 19th century in Scotland and Germany. Though the Triticale is a cross between wheat and rye, it is self-pollinating (similar to wheat) and not cross-pollinating (like rye).

149. Which one of the following is a man-made cereal, not found in nature?

(a) Dwarf wheat

(b) Hybrid maize

(c) Triticale

(d) Soybean

I.A.S. (Pre) 1993

Ans. (c)

See the explanation of above question.

150. In the nature, which of the following is/are most likely to be found surviving on a surface without soil?

1. Fern

2. Lichen

3. Moss

4. Mushroom

Select the correct answer using the code given below.

(a) 1 and 4 only

(b) 2 only

(c) 2 and 3

(d) 1, 3 and 4

I.A.S. (Pre) 2021

Ans. (c)

Land without soil can support only very small primitive plants such as lichens and mosses that can cling onto a rock surface. Lichens are generally found on substrates like rock, tree bark, soil, peat, houses, tombstones, old farm equipments etc. They can grow on wood, glass, metal, plastic, and even cloth and have no need of roots. Mosses are non-vascular plants and don't have true roots. They have a rhizoid multi-cell anchoring structure that is used to climb and grip over rocky surfaces. They thrive in moist and shady areas and do not need soil to survive. Hence, option (c) is the correct answer.

151. Read the given statements—

Assertion (A) : The root systems of desert plants are longer.

Reason (R) : High temperature of soil encourages the growth of roots.

Referencing above, which one is the right answer?

Code :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.

U.P. Lower Sub. (Pre) 1998

Ans. (c)

Desert plants have the capability to complete their life-cycle even in a dry atmosphere. The root system of these plants become longer in search of water from the soil and their stems are small, adapted and underground sometimes. Plant varieties that have adapted by developing very long roots are collectively known as phreatophytes.

152. Desert plants have longer root system because :

(a) High temperature of the soil stimulates the growth of roots

(b) Roots grow in search of water

(c) Soil does not contain water, hence becomes compact and presses the roots to grow longer

(d) Roots moves opposite to the solar heat

U.P.P.C.S. (Pre) 1996

Ans. (b)

See the explanation of above question.

153. Deserts have phreatophytes, i.e., plants that have :

(a) Succulent stem (100-200 mm thick)

(b) Lipid accumulation (20-30 mg) in leaves

(c) Long (20-30 ft) tap roots

(d) Small (2-3 mm) or spine-like leaves

R.A.S./R.T.S. (Re. Exam) (Pre) 2013

Ans. (c)

Phreatophyte is a deep-rooted plant that obtains a significant portion of water that it needs from the phreatic zone or the capillary fringe above the phreatic zone. These plants maintain water status which is largely independent of soil water. Their long tap roots (up to around 25-30 metres length) reaches underground water level to take water.

154. Plants adapted to dry climate are known as :

- (a) Epiphyte
- (b) Xerophyte
- (c) Hydrophytes
- (d) Mean plants

R.A.S./R.T.S. (Pre) 1993

Ans. (b)

A xerophyte is a species of plant that has adapted to survive in a dry environment. Cactus, opuntia, madar, nerium are the major plants of this type.

155. Desert plants are generally –

- (a) Viviparous
- (b) Succulent
- (c) Herbaceous
- (d) Heterophyllous

U.P. Lower Sub. (Pre) 2013

Ans. (b)

A succulent is a plant that stores water for times when water is not available to it. Succulent plants are generally found in arid environments such as deserts and semi-deserts.

156. Epiphytes are plants which depend on other plants for :

- (a) Food
- (b) Mechanical support
- (c) Shade
- (d) water

I.A.S. (Pre) 2001

Ans. (b)

An epiphyte is a plant that grows harmlessly upon another plant (such as a tree), fulfill moisture and nutrients from the air, rain and sometimes from debris accumulating around it, instead of the structure to which it is fastened. Epiphytic organisms usually derive only physical support and not nutrition from their host, though they may sometimes damage the host.

157. Plants which grow on saline soils are :

- (a) Xerophytes
- (b) Hydrophytes
- (c) Halophytes
- (d) Succulents

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Hydrophytes or aquatic plants are plants that have adapted to live in an aquatic environment. Plants growing on saline soils or water are called halophytes (salt-loving plants). They grow in saline semi-deserts, mangroove swamps, marshes and sloughs and seashores.

158. What are salt-loving plants called?

- (a) Xerophytes
- (b) Mesophytes
- (c) Glycophytes
- (d) Halophytes
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

See the explanation of above question.

159. A hydrophyte is :

- (a) A marine animal
- (b) An aquatic plant
- (c) A plant disease
- (d) A rootless plant

U.P.P.C.S. (Pre) 1994

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

See the explanation of above question.

160. Plants growing on rocks are called :

- (a) Epiphytes
- (b) Halophytes
- (c) Xerophytes
- (d) Lithophytes

R.A.S./R.T.S.(Pre) 2010

Ans. (d)

Lithophytes are plants that grow in or on rocks. Those that grow on rocks are also known as epipetric or epilithic plants. Lithophytes feed off nutrients from rain water and nearby decaying plants, including their own dead tissues.

161. Which one of the following pairs is not correctly matched?

- (a) Cuscuta – stem parasite
- (b) Chandan – root parasite
- (c) Orchids – epiphyte
- (d) Nepenthes – hydrophyte

U.P. R.O./A.R.O. (Pre) 2017

Ans. (d)

Nepenthes are not hydrophytes. They are tropical pitcher plants and a genus of carnivorous plant. Other pairs are correctly matched.

162. Consider the following statements :

1. Weeping jelly is due to an excess of acid.
2. Biofertilizer most suited to sugarcane are rhizobium.
3. Plants grown in salt water are called thallophytes.
4. Dormancy breaking hormone is cytokinin.

Which of these statements are correct ?

- (a) Only 1 and 2
- (b) Only 2 and 3
- (c) Only 3 and 4
- (d) Only 4 and 1

U.P. Lower Sub. (Pre) 2013

Ans. (d)

Weeping jelly is most often caused by an excess of acid in the fruit or recipe in general. So statement 1 is correct. Acetobacter diazotrophicus is more suitable for sugarcane crop than other biofertilizers. Therefore statement 2 is incorrect. A halophyte is a plant that grows in waters of high salinity, therefore, statement 3 is also wrong. Dormancy in some seeds is broken by treating them with hormones such as gibberellins and cytokinin. So statement 4 is correct.

163. Plants and animals that live on land are called :

- (a) Abiotic (b) Aquatic habitats
(c) Terrestrial habitats (d) Biotic

U.P.P.C.S. (Pre) 2024

Ans. (c)

Plants and animals that live on land are called terrestrial organisms, and their habitats are known as terrestrial habitats. These are the natural environments where terrestrial organisms live, providing them with the necessary resources for survival, such as food, water, shelter, and space. These include diverse environments like forests, grasslands, deserts and mountains.

164. Work as intermediaries between biotic and abiotic components :

- (a) Parasite (b) Decomposers
(c) Producers (d) Consumers

U.P. R.O./A.R.O. (Mains) 2017

Ans. (c)

Producers or autotrophs are the intermediaries between Biotic and abiotic components of the environment.

165. The process of breaking down organic matter by microorganisms is called :

- (a) Respiration (b) Photosynthesis
(c) Decomposition (d) Fermentation

70th B.P.S.C. (Pre) 2024

Ans. (c)

The process of breaking down organic matter by microorganisms is called decomposition. This is a natural process where microorganisms, like bacteria and fungi, break down complex organic substances (like dead plants and animals) into simpler inorganic substances. Decomposers are the microorganisms (bacteria, fungi, etc.) that carry out the decomposition process. The end products of decomposition include carbon dioxide, water, and nutrients, which are then released back into the environment and can be utilized by other organisms.

166. Which of the following is not correctly matched :

- (a) Autotroph – Producer
(b) Heterotroph – Consumer
(c) Saprotroph – Decomposer
(d) Herbivore – Secondary consumer

U.P.P.C.S. (Pre) 2000

Ans. (d)

Autotrophs are the producers in a food chain such as plants on land or algae in water, in contrast to heterotrophs as consumers of autotrophs. Saprotroph, a term used for organisms which obtain nutrients from dead organic matter (this term commonly applies to fungi). A herbivore is an

animal anatomically and physiologically adapted to eating plant material (for example foliage) for the main component of its diet. More generally, herbivores feed on autotrophs and are known as primary consumers. Thus, option (d) is not correctly matched.

167. Trophic level-I includes :

- (a) Herbivorous animals (b) Carnivorous animals
(c) Omnivorous animals (d) Green plants

M.P. P.C.S. (Pre) 2016

Ans. (d)

Trophic level-I includes all autotrophs which synthesize their food by themselves. All green plants belong to this trophic level. Organisms of trophic level-I usually utilize the radiant energy of the sun to synthesize their organic molecules.

168. Consider the following kinds of organisms :

1. Copepods 2. Cyanobacteria
3. Diatoms 4. Foraminifera

Which of the above are primary producers in the food chains of oceans?

- (a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4

I.A.S. (Pre) 2021

Ans. (b)

Copepods are a group of small crustaceans found in nearly every freshwater and saltwater habitat. Copepods are major primary consumers and secondary producers in the World Ocean. Most free-living copepods feed directly on phytoplankton, catching cells individually. Some of the larger species are predators of their smaller relatives. Many benthic copepods eat organic detritus or the bacteria that grow in it. Many copepods (e.g., fish lice like the Siphonostomatoida) are parasites, and feed on their host organisms.

Cyanobacteria, also called bluegreen algae, are microscopic organisms found naturally in all types of water. Cyanobacteria are important primary producers and form a part of the phytoplankton. They use photosynthetic pigments, such as carotenoids, phycobilins, and various forms of chlorophyll, which absorb energy from light.

Diatom refers to any member of a large group comprising several genera of algae, specifically microalgae, found in the oceans, waterways and soils of the world. They are especially important in oceans, where they contribute an estimated 45% of the total oceanic primary production of organic material.

Foraminifera are amoeba-like, single-celled protists (very simple micro-organisms). They are heterotrophic organisms which consumes smaller organisms like bacteria and organic matter. They are characterized by streaming granular ectoplasm for catching food and other uses; and commonly an external shell (called a "test") of diverse forms and materials.

169. Fishes survive under the frozen lake, because –

- (a) Fishes have hot blood
- (b) Fishes are in hibernation under it
- (c) Water does not freeze near bottom
- (d) Ice is the conductor of heat.

U.P.P.S.C. (GIC) 2010

Ans. (c)

Lakes and rivers are frozen when the outer temperature decreases below the freezing point in winters. But only the upper layer freezes in this condition. Water remains in liquid form below the upper layer. So fishes and other aquatic creatures can survive even in a frozen lake.

170. After bringing out of water fishes die, because –

- (a) They get excess quantity of oxygen
- (b) Their body temperature increases
- (c) They cannot breathe
- (d) They can not move in water

39th B.P.S.C. (Pre) 1994

Ans. (c)

Fishes respire with the help of their gills. Gills can readily absorb the oxygen dissolved in water. When fishes are taken out of water, the supply of oxygen to the fishes is cut as their gills cannot absorb the oxygen present in the atmosphere. Hence, they die after sometime.

171. Fish generally breathe through –

- (a) Skin
- (b) Nose
- (c) Gills
- (d) Fins

Uttarakhand P.C.S. (Mains) 2006

Ans. (c)

See the explanation of above question.

172. Redness in apple is due to –

- (a) Anthocyanin
- (b) Lycopene
- (c) Carotene
- (d) Xanthophyll

U.P.P.C.S. (Mains) 2002

Ans. (a)

Red apples get their colour from anthocyanin pigment.

173. Which of the following accounts for the red colour of tomato?

- (a) Capsaicin
- (b) Carotene
- (c) Anthocyanin
- (d) Lycopene

U.P. P.C.S. (Mains) 2016

Ans. (d)

The red colour of tomato (*Lycopersicon esculentum*) fruit is due to the carotenoid pigment lycopene. The red colour formation in the apple skin is due to the formation of anthocyanin pigments. The red-orange colour of carrot is due to the pigment β -Carotene.

174. Which one of the following is responsible for the red colour of tomato?

- (a) β -Carotene
- (b) Anthocyanin
- (c) Lycopene
- (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

See the explanation of above question.

175. Red colour in tomato is due to :

- (a) Capsaicin
- (b) Lycopene
- (c) Xanthophyll
- (d) None of the above

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

See the explanation of above question.

176. The yellow colour of papaya is due to -

- (a) Papain
- (b) Lycopene
- (c) Caricaxanthin
- (d) Carotene

U.P. Lower Sub. (Pre) 2015

Ans. (c)

The botanical name of papaya is 'Carica papaya'. It is rich in carotene, which is a precursor of vitamin A. The papain enzyme presented in papaya is helpful in digestion of protein. The yellow colour of papaya is due to presence of caricaxanthin.

177. Recently, our scientists have discovered a new and distinct species of banana plant which attains a height of about 11 metres and has orange-coloured fruit pulp. In which part of India has it been discovered?

- (a) Andaman Islands
- (b) Anaimalai Forests
- (c) Maikala Hills
- (d) Tropical rainforests of northeast

I.A.S. (Pre) 2016

Ans. (a)

Scientists at the Botanical Survey of India (BSI), have discovered a new species of banana from a remote tropical rain forest on the Little Andaman Islands. The species, *Musa indandamanensis*, was located about 16 km inside the Krishna- Nalah forest on the Island. The scientist who has made discovery described it as a distinct global species with unique green flower and fruit bunch lux thrice the size of a regular banana species. The new species is about 11 metres high, whereas the usual banana species is about three to four metres high. The fruit lux of the new species is about one metre which are thrice the size of the regular species.

178. Which of the following groups of living beings belong to the same species?

- (a) Chinese, American, Indians and Black Africans
- (b) Tigers, Lions, and Cats

- (c) Pigeons, Doves, and Partridges
(d) Lizards, Crocodiles, and Snakes

R.A.S./R.T.S.(Pre) 2003

Ans. (a)

Chinese, American, Indians and Black Africans groups of living beings belong to the same species. Homo sapiens is the binomial nomenclature (also known as the scientific name) for the human species. While the animals in other options are related to different species.

179. Fungi are plants that lack :

- (a) oxygen
(b) carbon dioxide
(c) chlorophyll
(d) sunlight
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (c)

A fungus (Plural : fungi) is any member of the group of eukaryotic organisms that includes microorganisms such as yeasts and molds, as well as the more familiar mushrooms. These organisms are classified as a kingdom, separately from the other eukaryotic kingdoms (earlier they were considered plants). Fungi lack chlorophyll so they do not photosynthesize.

180. Yeast is an example of :

- (a) Bacteria
(b) Fungus
(c) Virus
(d) Algae

U.P. Lower Sub. (Pre) 2013

Ans. (b)

Yeasts, moulds (molds) and mushrooms are examples of fungi. Fungi is the plural word for fungus. Fungi lack chlorophyll and vascular bundle.

181. Yeast and mushrooms are :

- (a) Algae
(b) Gymnosperm
(c) Fungi
(d) Tuberous Roots

Uttarakhand P.C.S. (Pre) 2003

Ans. (c)

See the explanation of above question.

182. What is a mushroom?

- (a) Fungi
(b) Plant
(c) Animal
(d) Bacteria

Chhattisgarh P.C.S. (Pre) 2011

Ans. (a)

See the explanation of above question.

183. Mushroom used in for making juicy vegetable is a –

- (a) Fungus
(b) Alga
(c) Green vegetable
(d) Flesh of animal

U.P.P.C.S. (Mains) 2008

Ans. (a)

See the explanation of above question.

184. With reference to 'Gucchi' sometimes mentioned in the news, consider the following statements :

1. It is a fungus.
2. It grows in some Himalayan forest areas.
3. It is commercially cultivated in the Himalayan foothills of north-eastern India.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 3 only
(c) 1 and 2
(d) 2 and 3

I.A.S. (Pre) 2022

Ans. (c)

Gucchi mushroom (morels) is a species of fungus in the family Morchellaceae of the Ascomycota. They are pale yellow in colour with large pits and ridges on the surface of the cap, raised on a large white stem. Guccchi is known to be one of the most expensive mushrooms. Apart from its flavour, guccchi also owes its high price to the challenges in cultivation. These mushrooms cannot be cultivated commercially and grow in Himalayan conifer forests across temperate regions, and the foothills in Himachal Pradesh, Uttarakhand, and Jammu and Kashmir. And it takes months for villagers to collect enough of these mushrooms, dry them and bring them to the market.

185. Which one of the following shows a unique relationship with an insect that has coevolved with it and that is the only insect that can pollinate this tree?

- (a) Fig
(b) Mahua
(c) Sandalwood
(d) Silk cotton

I.A.S. (Pre) 2024

Ans. (a)

Plant-animal interactions often involve co-evolution of the mutualists, that is, the evolutions of the flower and its pollinator species are tightly linked with one another. In many species of fig trees, there is a tight one-to-one relationship with the pollinator species of wasp. It means that a given fig species can be pollinated only by its 'partner' wasp species and no other species. The female wasp uses the fruit not only as an oviposition (egg-laying) site but uses the developing seeds within the fruit for nourishing its larvae. The wasp pollinates the fig inflorescence while searching for suitable egg-laying sites. In return for the favour of pollination the fig offers the wasp some of its developing seeds, as food for the developing wasp larvae.

186. Which of the following have species that can establish symbiotic relationship with other organisms?

1. Cnidarians
2. Fungi
3. Protozoa

Select the correct answer using the code given below.

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2021

Ans. (d)

Cnidaria is a phylum under Kingdom Animalia containing over 11,000 species of aquatic animals found both in freshwater and marine environments. Mostly marine animals, the cnidarians include the corals, hydras, jellyfish, Portuguese men-of-war, sea anemones, sea pens, sea whips, and sea fans. The symbiosis between cnidarians (e.g., corals or sea anemones) and intracellular dinoflagellate algae of the genus Symbiodinium is of immense ecological importance. In particular, this symbiosis promotes the growth and survival of coral reefs in nutrient-poor tropical waters; indeed, coral reefs could not exist without this symbiosis.

Fungi have several symbiotic (mutualistic) relationships with other organisms. In mutualism, both organisms benefit from the relationship. Two common mutualistic relationships involving fungi are mycorrhiza and lichen. Mycorrhiza are symbiotic relationships that form between fungi and vascular plants. The fungi colonize the root system of a host plant, providing increased water and nutrient absorption capabilities while the plant provides the fungus with carbohydrates formed from photosynthesis. A lichen is an organism that results from a mutualistic relationship between a fungus and a photosynthetic organism. The other organism is usually a cyanobacterium or green algae. The fungus grows around the bacterial or algal cells. The fungus benefits from the constant supply of food produced by the photosynthesizer. The photosynthesizer benefits from the water and nutrients absorbed by the fungus.

Symbiosis in protozoa mostly represents a close mutualistic association between a protozoan and unicellular symbionts (bacteria, cyanobacteria or/and unicellular algae) or protozoans and a multicellular organism (ruminants, lower termites, wood-eating cockroaches, plants).

187. Which one of the following is a useful functional association between fungi and the roots of higher plants ?

- (a) Biofertilizer (b) Coralloid root

- (c) Lichens (d) Mycorrhiza

I.A.S. (Pre) 1999

Ans. (d)

See the explanation of above question.

188. 'Mycorrhiza' is a symbiotic association between :

- (a) Algae and plants
(b) Algae and fungi
(c) Fungi and plants
(d) Blue-green algae and fungi

U.P. B.E.O. (Pre) 2019

Ans. (c)

See the explanation of above question.

189. Lichens, which are capable of initiating ecological succession even on a bare rock, are actually a symbiotic association of –

- (a) Algae and bacteria (b) Algae and fungi
(c) Bacteria and fungi (d) Fungi and mosses

I.A.S. (Pre) 2014

Ans. (b)

Lichens are association of two organisms that grow together in a symbiotic relationship . Lichens are a partnership between fungi and algae.

190. Lichen are the complex creature, made from–

- (a) Fungi and Bacteria (b) Fungi and Algae
(c) Fungi and Bryophyta (d) Algae and Bacteria

R.A.S./R.T.S. (Pre) 1997

Ans. (b)

See the explanation of above question.

191. The interaction between algae and fungi to form lichen is called :

- (a) parasitism (b) mutualism
(c) commensalism (d) conversion
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

The interaction between algae and fungi to form lichen is called mutualism. Symbiosis of algae and fungi in lichens is the mutually helpful symbiotic relationship between them. The algae or cyanobacteria (chlorophyll – containing partner) benefit their fungal partner by producing organic compounds through photosynthesis. In return, the fungal partner benefits